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Advanced glaucoma detection in obscured retinal images via refined blur map and optic disc/cup segmentation

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ABSTRACT

Glaucoma is the leading cause of permanent blindness, mostly because of delayed diagnosis and the poor visual cues observed in fundus imaging. Automated screening methods sometimes exhibit poor performance in low-quality or fuzzy retinal images, leading to inaccurate optic disc and cup segmentation and unpredictable cup-to-disc ratio (CDR) computation. To overcome these limitations, this study proposes a blur-aware glaucoma detection system that combines strong deep learning-based segmentation and classification with image quality augmentation. First, a focus measure based on Laplacian variance is used to build a blur map, which is then used to selectively supplement the damaged retinal regions. For optic disc and cup segmentation, an Energy Mask R-CNN is then used, where a novel energy-based loss function enhances robustness against noise and reinforces boundary preservation. A cyclic code encoding approach is used to further process the segmentation outputs, guaranteeing persistent representation of morphological and CDR information. Finally, images are sorted into two groups: healthy or showing signs of glaucoma, using a special type of network called a Cyclic Code-based Long Short-Term Memory (CC-LSTM) network. The proposed system achieved a 98% accuracy rate, along with improved results in terms of Dice coefficient and area under the curve (AUC), as illustrated through experiments on standard datasets. It performed exceptionally well with unclear or low-quality images. These results indicate that the tool can be a reliable choice for broader glaucoma screening.

1. Introduction

Glaucoma is a long-term health issue that gradually damages the optic nerve and can result in permanent loss of vision. It is one of the main causes of blindness globally [1]. Many individuals are unaware they have glaucoma at an early stage because the condition often does not present with obvious signs [2]. As a result, a lot of people delay getting help until they start experiencing major vision issues. Research predicts that up to 110 million people worldwide may have glaucoma by 2040, highlighting the importance of improving ways to detect and diagnose the condition. Glaucoma is usually diagnosed using tests like measuring eye pressure, checking the visual field, and examining the optic nerve head [3,4].

Among these, the cup-to-disc ratio (CDR) is the most important. CDR measures how large the optic cup is compared to the optic disc in terms of height. This measurement plays a key role in making an accurate diagnosis because it is closely linked to damage in the optic nerve. Even though identifying the optic disc (OD) and optic cup (OC) in fundus images is essential for diagnosing glaucoma, doing this accurately is still a big challenge [5]. The OC often appears brighter than the surrounding tissues, and its edges are not clearly defined, which makes it difficult to determine its boundaries, especially in lower quality images [6]. When ophthalmologists perform this task manually, they rely on their individual judgment, which can be time-consuming and heavily dependent on their experience. As a result, different physician might produce varying results [7,8]. Fig. 1 illustrates the structural differences between eyes with glaucoma and those that are healthy. It also clearly highlights the noticeable increase in the cup-to-disc ratio (CDR). These findings stress the importance of reliable and automated tools to enhance the standardization of glaucoma diagnosis. By minimizing errors from human factors and ensuring more consistent results, automated technology can help save time and effort in clinical settings.

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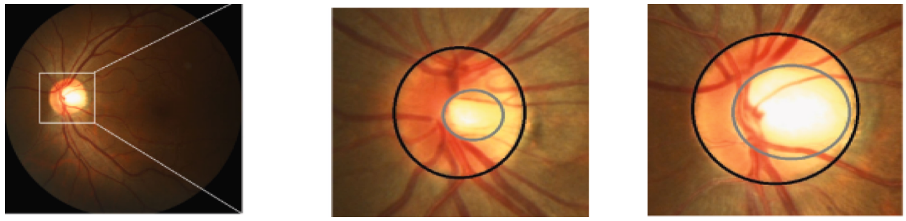

Fig. 1. Schematic representation of OD and OC boundaries in fundus imaging. The black line delineates the OD margin, while the grey line demarcates the OC region, illustrating the characteristic increase in CDR observed in glaucoma cases compared to normal ocular anatomy.

Recent advances in medical image processing and artificial intelligence (AI) have transformed the way eye diseases are diagnosed [9,10]. AI techniques have proven to be highly effective in identifying blood vessels, detecting diabetic retinopathy, and analyzing retinal images [11,12]. When dealing with a large set of images, these methods help increase the accuracy of diagnosis, minimize errors caused by human error, and provide consistent and reliable results each time. They are also good at managing big sets of medical data, which helps in early disease detection and screening many people [13,14]. Compared to traditional methods, researchers can now create more accurate tools for detecting glaucoma using AI. In the end, AI-based methods offer a promising way to develop more efficient and easy-to-use glaucoma screening programs, which can improve patient care and help prevent blindness [15]. Previous studies have explored ways to detect blur and separate the optic disc and cup in glaucoma patients, but our approach introduces some fresh ideas. We use a blur map that adapts to the image's structure, which helps bring out the blurry areas while keeping the clear parts untouched. This makes it easier to find useful features. We also use an Energy Mask R-CNN with a loss function that helps keep edges sharp, allowing for accurate segmentation of the optic disc and cup, even in unclear or noisy images. Another important part of our method is an LSTM classifier based on cyclic codes, which helps recognize patterns in retinal images that relate to rotation and structure. This improves the reliability of glaucoma classification. By combining all these elements, our system works better than existing methods, especially when dealing with blurry or low-quality retinal images, making it a strong option for real-world glaucoma screening.

The main contribution of this research is,

- Utilizes a spatially adaptive blur map to enhance the clarity of blurry areas in the retina, making it easier to identify and extract key features.
- It's recommended to use an Energy Mask R-CNN model along with a loss function that helps keep edges clear, leading to very accurate object detection and segmentation.
- Utilizes a cyclic code-based LSTM to effectively classify glaucoma, which helps in recognizing patterns that remain consistent even when rotated and captures how different parts of the structure relate to each other.
- Provides a complete solution that works from start to finish, making retinal images clearer and more reliable, even when the images are of poor quality.

This document is organized in the following manner: Section 2 covers related work, while Section 3 presents the proposed methodology involving Energy Mask R-CNN and classification using a Cyclic Code-based LSTM. Section 4 consists of the Results and Discussion, which assesses the system, and Section 5 wraps up with the Conclusion.

2. Related work

Kim, et.al [16] created a deep learning system that uses Mask R-CNN to identify the optic disc and MAPNet to separate the disc from the cup. The system automatically checks important clinical measurements like the cup-to-disc ratio and rim-to-disc ratio. When tested on standard datasets, it demonstrated high accuracy in detecting these structures, showing that it can be a reliable tool for assessing glaucoma in real-world medical settings. By grouping the steps of detecting and separating the optic structures into one process, the system works more efficiently and needs less manual effort. Their results indicate that this method has great potential for use in automated glaucoma screening in actual healthcare environments.

Natarajan, et.al [17] created a two-step deep learning model named UNet-SNet, which is used to identify the optic disc and detect glaucoma. They trained the model using datasets like RIGA and RIM-ONE, and it achieved segmentation accuracy higher than 97% and glaucoma detection accuracy over 99%. The method relies on thorough feature extraction and strong classification skills. The model's great performance highlights its value in automated screening. It pays attention to both precise results and efficient use of computer resources. The research demonstrates that pairing segmentation with classification can enhance the accuracy of diagnoses.

Singh, et.al [18] created two new techniques for choosing important features, named BA-BCS and BCS-PSO. These methods mix different types of advanced algorithms to make diagnosing eye diseases more accurate. They tested these techniques on standard datasets and found that they could reach an accuracy of up to 98.95%. The combination of these methods works well because it keeps the diagnosis accurate while also speeding up the process. It helps eliminate extra information from the data, which makes the classification more efficient. The results suggest that this method has a good potential to be used in bigger, automated systems for detecting conditions like glaucoma. The way it is structured makes it easy to integrate into the current practices used in eye clinics. Overall, this offers a reliable and better way to analyze data related to eye health.

Puchaicela-Lozano, et.al [19] developed a combined method for detecting glaucoma that uses ResNet-50 R-CNN to examine fundus images, along with a gradient-based method to estimate the cup-to-disc ratio. When tested on the ORIGA and ACRIMA datasets, their method achieved an accuracy of 95%. Their approach combines deep learning with important clinical features related to the cup-to-disc ratio, which helps make the diagnosis more clear. The method still performs well even when the images are not very sharp. By mixing feature-based methods with deep learning, the approach becomes more reliable.This system supports automatic glaucoma screening across different medical environments. It also shows how merging data-driven strategies with clinical expertise can be beneficial.

Almustofa, et.al [20] improved a U-Net model to identify and separate the optic disc and cup in retinal images without needing labeled data. They tested their model on two datasets named Drishti-GS and REFUGE, and it was effective at recognizing both the disc and the cup. Because the model doesn't require labeled data, it can be used more easily in different situations. The model performs well even when images are captured under different conditions. It is also simple to use in a clinical setting with minimal setup. The model's high accuracy supports doctors in more reliable glaucoma assessment. This approach offers a

Table 1
Evaluate the analysis of existing work.

Author	Method	Objectives	Dataset	Advantages	Disadvantage
Kim et al. [16]	Automated deep learning-based segmentation using Mask R-CNN and MAPNet	To assess crucial clinical measures and separate the optic disc and cup from fundus images	–	Effective clinical metric estimation with low average errors.	Requires an extensive dataset for both validation and training
Natarajan et al. [17]	UNet-SNet: A lightweight two-stage network using UNet for OD segmentation and SqueezeNet for glaucoma classification	To identify glaucoma using optic disc (OD) segmentation and fundus image classification as either normal or glaucomatous	RIGA, RIM-ONEv2, ACRIMA, Drishti-GS1, RIM-ONEv1	High segmentation accuracy (97.84% on RIGA, 99.85% on RIM-ONEv2).	Requires multiple datasets for training and validation.
Singh et al. [18]	Binary Cuckoo Search (BCS), Particle Swarm Optimization (PSO), and Bat Algorithm (BA) are used in two-layered feature selection techniques (BA-BCS and BCS-PSO).	To optimize feature selection for improving classification accuracy in glaucoma detection	ORIGA, REFUGE	High classification accuracy (up to 98.95%).	Computational complexity due to multiple optimization techniques.
Puchaicela-Lozano et al. [19]	A hybrid approach integrating cup-to-disc segmentation with a pre-trained R-CNN	To detect glaucoma in fundus images by combining deep learning with traditional segmentation techniques	ORIGA, CRIME	Achieves 95% accuracy in glaucoma detection.	Requires high-quality fundus images for optimal performance.
Almustofa et al. [20]	Enhanced U-Net Framework for Joint Optic Disc/Cup Semantic Segmentation with Unsupervised Disc Localization	To enhance optic disc localization and segmentation in retinal images using an improved U-Net	Drishti-GS, Refuge	High localization accuracy (99.99 ± 0.12% of optic disc pixels in ROIs).	Performance may vary across different datasets.
Mangipudi et al. [21]	Modified ground truth from several expert annotations is used for sophisticated deep learning-based optic disc and cup segmentation.	To incorporate fused expert marks to increase the precision and resilience of optic disc and cup segmentation	DRISHTI-GS, DRIONSDB, RIM-ONE v3	High segmentation accuracy with overlapping scores up to 98.42%.	Computationally intensive due to deep learning model training.

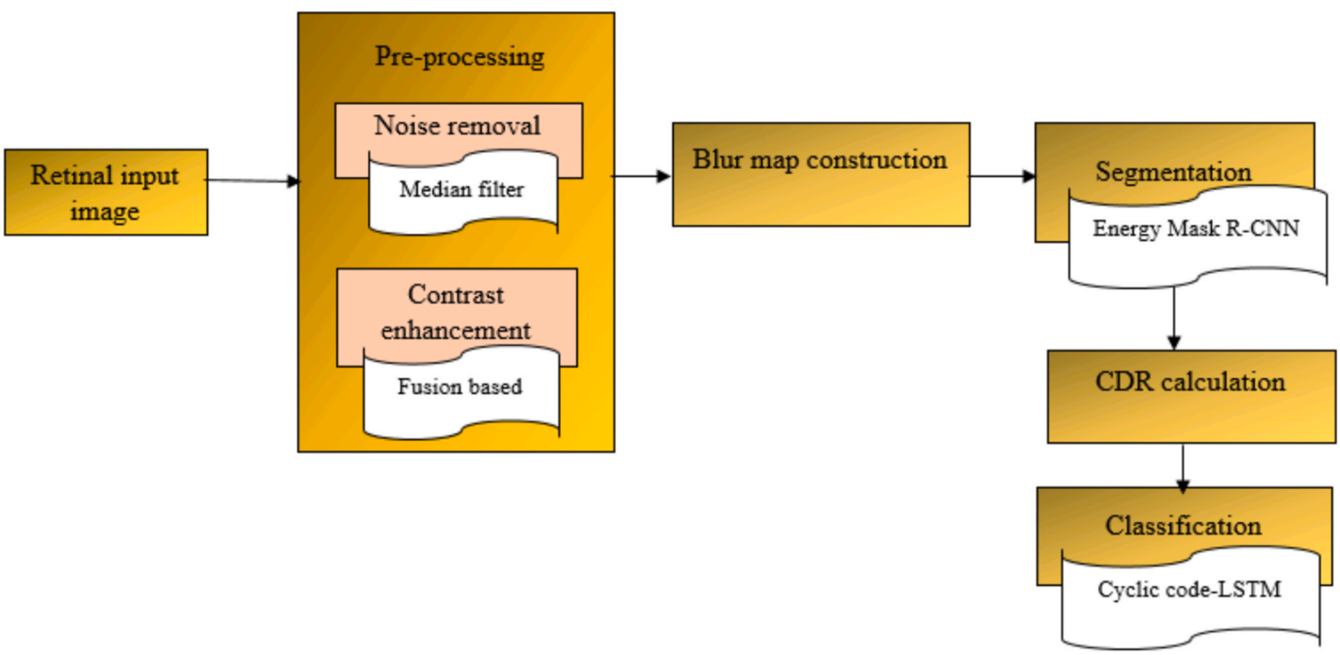

Fig. 2. The proposed diagram for Glaucoma Detection.

helpful way to automatically analyze retinal images.

Mangipudi, et.al [21] developed a better deep learning model that uses training data enhanced by expert input to segment the optic disc and cup. When tested on datasets like DRISHTI-GS, DRIONS-DB, and RIM-ONE, the model scored over 94% in overlap. Using data that includes expert input helped the model perform better in different situations and made its results more reliable. The method is good at balancing accuracy with efficiency in terms of computation. The evaluation shows that the model works well across various datasets. This method can also be applied in real clinical environments. It demonstrates how merging expert insights with deep learning can significantly aid in identifying glaucoma. Table 1 evaluate the analysis of existing work.

2.1. Problem statement

- Traditional glaucoma diagnosis methods rely on high-quality retinal images, but blurring from poor imaging settings, motion artifacts, and ocular media opacities can drastically reduce image quality. This decline in quality obstructs the precise segmentation of the optic disc and cup, which can lead to unreliable cup-to-disc ratio (CDR) values and the possibility of misdiagnosis.
- Standard segmentation techniques struggle to accurately identify the borders of the optic disc and cup in unclear retinal images. The absence of robust segmentation models results in errors in CDR calculations, which diminishes the effectiveness of automated glaucoma detection, particularly in clinical settings where image quality is inconsistent.

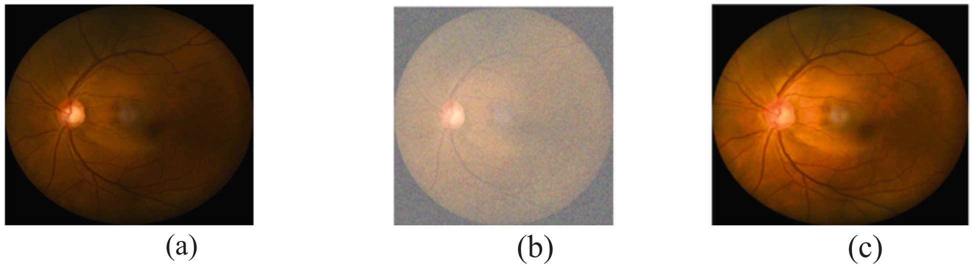

Fig. 3. Sample retinal images for (a) Original image, (b) Noise Image, (c) Contrast.

- Existing methods for assessing blur fail to take into account the spatial variability of defocus in retinal images. Current techniques tend to either apply uniform deblurring or neglect the enhancement of vital retinal regions, resulting in subpar image restoration and reduced accuracy in extracting features essential for evaluating glaucoma.

3. Proposed method

The proposed work focuses on an automated glaucoma detection method that leverages retinal pictures, both blurred and unblurred. Preprocessing techniques are first used to improve image quality and remove noise with a Gaussian filter, followed by fusion-based contrast enhancement. A blur map is then created to analyze the sharpness of the image and aid in subsequent processing. The Energy Mask Region-CNN model is used for the segmentation of the optic disc and cup, ensuring accurate identification of these vital regions. The Cup-to-Disc Ratio (CDR), which is a key indicator in glaucoma diagnosis, is subsequently computed. A Cyclic Code-based Long Short-Term Memory (LSTM) network is employed for classification to effectively differentiate between glaucomatous and non-glaucomatous cases. Ultimately, this technology aids healthcare professionals in decision-making by offering a thorough risk evaluation and visualization of the segmented areas. Fig. 2 illustrates the block diagram of the proposed model.

3.1. Data pre-processing

Initially, the retinal picture (Blurred/Unblurred) is noise-removed using techniques such as median filtering to reduce artifacts while keeping structural features. Next, contrast enhancement techniques, such as fusion-based approaches, enhance the visibility of retinal features. Fig. 3 shows retinal pictures sampled in (a) the original image, (b) the noise version, and (c) the contrast version.

3.1.1. Noise removal

Noise in retinal images is frequently caused by compression artifacts, sensor limitations, or acquisition conditions. A common technique for denoising while maintaining important structures is median filtering. The median of the surrounding pixels within a specified window is used to replace each pixel value in the median filter. In terms of mathematics, this is provided by:

$$I'(x,y) = median\{I(i,j)|(i,j) \in N(x,y)\} \tag{1}$$

Where $I(x,y)$ is the original pixel intensity, $I'(x,y)$ is the filtered pixel value, and $N(x,y)$ represents the neighborhood pixels.

3.1.2. Contrast enhancement

After eliminating noise, contrast enhancement is applied to enhance the visibility of retinal structures, which is essential for medical evaluation. Wavelet transform fusion serves as an effective contrast enhancement method that breaks down an image into various frequency components and enhances specific features selectively. This Discrete wavelet transform is commonly used, where the original image $I(x,y)$ is decomposed into approximation A and detail coefficients (H,V,D):

$$I(x,y) \rightarrow \{A(x,y), H(x,y), V(x,y), D(x,y)\} \tag{2}$$

Where $A(x,y)$ denotes the low-frequency approximation and $H(x,y)$, $V(x,y)$, $D(x,y)$ records high-frequency features in the horizontal, vertical, and diagonal directions.

For fusion, many improved versions of the image are processed using the wavelet transform, and the fused coefficients are generated using a selection rule, such as maximum selection or weighted averaging:

$$C_f = F(C_1, C_2, \ldots\ldots, C_n) \tag{3}$$

Where C_f are the fused coefficients, and F is the fusion rule applied to multiple enhanced images $C_1, C_2, \ldots\ldots, C_n$. The inverse wavelet transform is then applied to reconstruct the enhanced image.

$$I_f(x,y) = IDWT(A_f, H_f, V_f, D_f) \tag{4}$$

This method preserves high-frequency details like blood vessels and microaneurysms while increasing global contrast, making the retinal image more appropriate for further study and medical diagnosis.

3.2. Refined blur map

Produce a per-pixel sharpness map $B_r(x,y) \in [0,\ 1]$. This map is used to (i) selectively enhance blurred regions, (ii) be supplied as an auxiliary input channel to the segmentation network, and (iii) derive pixel-wise loss weights to emphasize blurred regions during training.

3.2.1. Laplacian response

Compute the discrete Laplacian with kernel $[MICK]_{\nabla^2}$:

$$[MICK]_{\nabla^2} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix} L(x,y) = ([MICK]_{\nabla^2} * I_{enh})\ (x,y) \tag{5}$$

Edge magnitude:

$$R(x,y) = |L(x,y)| \tag{6}$$

3.2.2. Local variance of Laplacian

For a neighborhood $N_w(x,y)$ of size $w \times w$:

$$\mu_L(x,y) = \frac{1}{|N_w|} \sum_{(p,q)\in N_w} L(p,q), \qquad V(x,y) = \frac{1}{|N_w|} \sum_{(p,q)\in N_w} (L(p,q) - \mu_L(x,y))^2 \tag{7}$$

3.2.3. Normalize &fuse

Min-max normalize with stability ε:

$$\widetilde{R}(x,y) = \frac{R(x,y) - \min R}{\max R - \min R + \varepsilon}, \quad \widetilde{V}(x,y) = \frac{V(x,y) - \min V}{\max V - \min V + \varepsilon} \tag{8}$$

Fuse:

$$S(x,y) = \alpha.\widetilde{R}(x,y) + (1-\alpha)\widetilde{V}(x,y), \qquad \alpha \in [0,1] \tag{9}$$

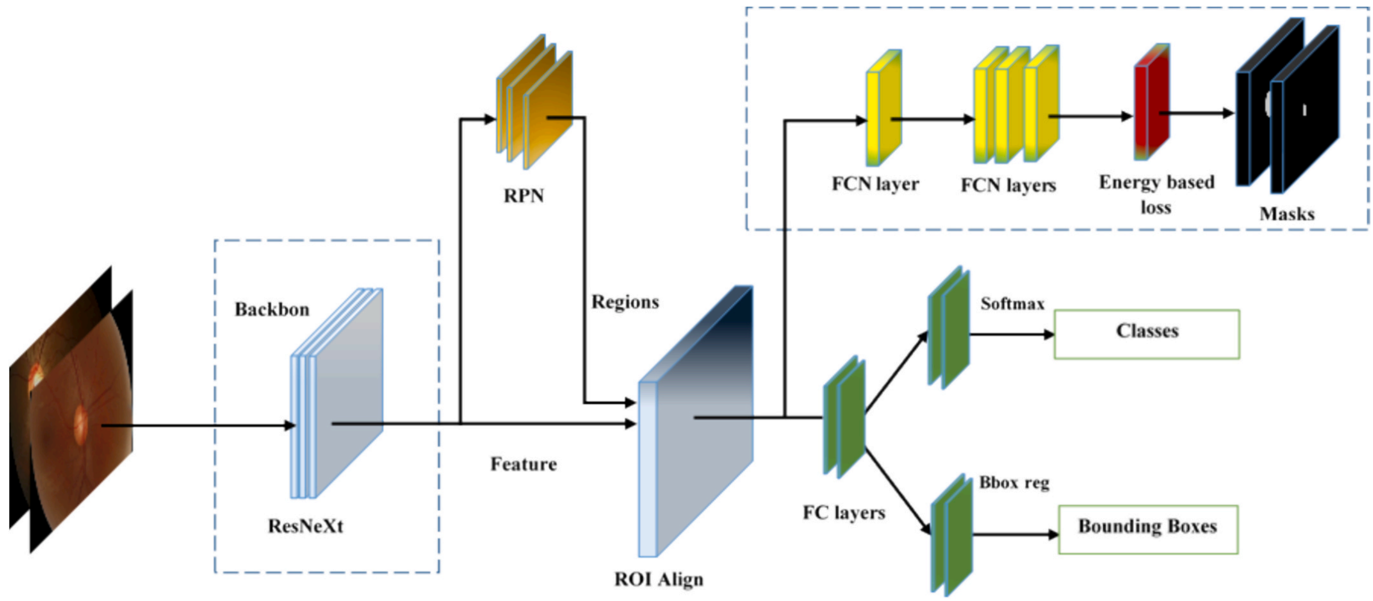

Fig. 4. Proposed Energy Mask R-CNN.

3.2.4. Spatial smoothing

Smooth with Gaussian G_σ:

$$B_r(x,y) = (G_\sigma * S)\ (x,y) \tag{10}$$

Re-normalize $B_r \in [0,1]$, optionally compute blur intensity map:

$$BlurMap(x,y) = 1 - B_r(x,y) \tag{11}$$

Binary blurred-region mask by threshold τ

$$M_{blur}(x,y) = 1\{1 - B_r(x,y) < \tau\} \tag{12}$$

3.2.5. Selective enhancement

Let $D(\cdot)$ be a deblurring operator. For reproducibility, we recommend unsharp masking:

$$I_{unsharp} = I_{enh} + k\left(I_{enh} - G_{\sigma_d} * I_{enh}\right), \qquad k \in [0.5,\ 1.5] \tag{13}$$

Blend original and deblurred:

$$I_{sel}(x,y) = B_r(x,y)\ I_{enh}(x,y) + (1 - B_r(x,y))\ D(I_{enh})(x,y) \tag{14}$$

This preserves naturally sharp areas while restoring blurred regions. Algorithm 1

Refined Blur Map construction

Input: I, W, σ, α, k, τ
Output: B_r, Blur Map I_{sel}
1. Compute Laplacian: $L = conv(I, k_{\nabla^2})$
2. $R = abs(L)$
3. For each pixel (x,y): compute μ_L and over $N_w(x,y)$
4. Normalize $R_n = normalize(R)$, $V_n = normalize(v)$
5. Fuse $S = \alpha * R_n + (1-\alpha) * V_n$
6. Smooth $B_r = gaussian\_filter(S, \sigma)$
7. Normalize B_r *to* $[0,1]$
8. Blur Map=1 $-B_r$
9. Compute Blur Mask via τ
10. Compute $I_{unsharp}$ using Eq. (13)
11. $I_{sel} = B_r * I + (1 - B_r) * I_{unsharp}$
12. Return B_r, Blur Mask, I_{sel}

3.3. Segmentation

The fundamental distinction between Energy Mask Region-CNN and regular Mask Region-CNN [21] is their approach to segmentation accuracy, particularly at object borders. Mask R-CNN enables instance segmentation by extending Faster R-CNN with a branch for pixel-wise mask prediction. However, because it relies on ordinary cross-entropy loss for mask generation, it frequently fails to achieve boundary precision, particularly in fuzzy or low-contrast images. In contrast, Energy Mask R-CNN improves the classic design by incorporating an energy-based loss function that focuses on fine-tuning border features. This energy-based loss reduces misclassifications at object edges while preserving fine structural details, yielding more accurate and precise segmentation results. This increase is notably useful in medical imaging jobs that need accurate boundary delineation for diagnostic accuracy, such as optic disc and cup segmentation in retinal images.

To extract OD and OC with accuracy, the suggested Energy Mask Region-CNN model is used. The existing Mask R-CNN [21] conducts pixel-wise segmentation but struggles with accuracy in hazy retinal pictures due to its reliance on standard feature extraction. To achieve this, a modified Mask R-CNN model with an energy-based loss function is used to improve boundary precision and segmentation accuracy. This enhancement leads to more accurate CDR calculations, which improves glaucoma diagnosis even in tough, hazy pictures. Fig. 4 shows the proposed Energy Mask R-CNN model.

3.3.1. Model Architecture

Energy Mask R-CNN is an improved variant of Mask R-CNN designed for medical image segmentation (OD/OC). It extends Faster R-CNN with an additional mask prediction branch while integrating an energy-based loss to enhance boundary precision.

a) Backbone Network: ResNet-50 with feature pyramid Network (FPN) for multiple scale feature extraction.

$$F = f_\theta(I) F \in R^{H\times W\times C} \tag{15}$$

Where I = Input retinal image, F=extracted feature map, f_θ=CNN transformation.

b) Region proposal Network:

Generates region proposal R, each scored with a sigmoid classifier

$$P(r) = \sigma(W_r \cdot F_r + b_r), \qquad r \in R \tag{16}$$

c) ROI Align Layer: Avoids spatial misalignment using bilinear interpolation to produce fixed-size feature maps:

Table 2
Parameters and values for Fine-Tuning Energy Mask R-CNN.

Parameter	Value/Range
Learning Rate	0.0001 to 0.001
Size of Batch	2 to 8
The number of epochs	10 to 50
Backbone	ResNet-50
Optimizer	Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$)
Weight Decay	0.0001 to 0.0005
Anchor Scales for RPN	[8,16,23]
Anchor Ratios for RPNs	[0.5, 1, 2]
RoI Align Pool Size	7x7
Mask Head Output Size	28x28
Energy-Based Loss Weight	$\lambda = 0.1$ to 1.0
Edge-Preserving Loss Weight	λ_edge = 0.1 to 1.0
Class Weights	[1.0, 2.0] (for OD and OC, respectively)
Early Stopping Patience	5 to 10 epochs
Gradient Clipping	1.0 to 5.0

$$F_{rol} = \sum_{i,j} w_{ij} . F\left(x_i, y_j\right) \tag{17}$$

d) Mask Head: Predicts pixel-wise segmentation masks at resolution 28×28

$$M\frown(x,y) = \sigma(W_m * F_{roi} + b_m) \tag{18}$$

3.3.2. Training strategy

The proposed Energy Mask R-CNN was trained using transfer learning with COCO pre-trained weights, which were fine-tuned on the retinal dataset to accelerate convergence. The dataset was split into 70% training, 15% validation, and 15% testing, with data augmentation techniques such as rotation, flipping, Gaussian blur, and brightness adjustments applied to improve generalization. The Adam optimizer ($\beta_1 = 0.9,\ \beta_2 = 0.999$) was used with an initial learning rate 1×10^{-4} that decayed gradually to stabilize training. Regularization was enforced through weight decay (0.0001–0.0005) and gradient clipping (1.0–5.0) to prevent overfitting and exploding gradients.

Training was conducted for 10–50 epochs with a small batch size (2–8), and early stopping (patience of 5–10 epochs) was employed to avoid unnecessary computation once validation Dice scores plateaued. The overall loss was a weighted sum of classifications, bounding box regression, Mask energy-based, and edge preserving losses, where λ and λ_{edge} were tuned in the range 0.1–1.0. This strategy ensured stable optimization while enhancing boundary sharpness, ultimately improving the segmentation accuracy of the OD or OC in retinal fundus images. Table 2 summarizes the parameters and values for the proposed Energy Mask R-CNN.

3.3.3. Loss functions

The proposed energy Mask R-CNN employs a multi-term composite loss function that balances classification, localization, segmentation accuracy, and boundary refinement. Each component is mathematically defined as follows:

a. Classification loss (L_{cls}).

The classification branch uses the cross-entropy loss to distinguish foreground from background:

$$L_{cls} = -\sum_i y_i \log(\widehat{y}_i) \tag{19}$$

Where y_i is the ground truth label, and $\widehat{y}_i$ is the predicted class probability.

b. Bounding Box regression

For precise localization of OD/OC regions, smooth L_1 loss is applied:

$$L_{box} = \sum_i Smooth\ L_1(b_i - \widehat{b}_i) \tag{20}$$

c. Mass Loss (L_{mark}).

To ensure accurate segmentation at the pixel level, binary cross-entropy is applied:

$$L_{mask} = -\sum_{p\in\Omega} (M_p\ Log(\widehat{M}_p) + (1 - M_p)\log(1 - \widehat{M}_p)\ ) \tag{21}$$

Where M_p and $\widehat{M}_p$ represent the ground truth and predicted mask values at pixel p.

d. Energy-Based loss (L_{energy}).

This term reduces false boundaries by penalizing energy deviations between predicted and ground truth masks:

$$L_{energy} = \|M - \widehat{M}\|_2^2 \tag{22}$$

Where M a $\widehat{M}$ re the ground truth and predicted masks, respectively.

e. Edge-preserving loss.

To refine optic disc and cup boundaries, an edge-preserving loss aligns the gradient of predicted and ground truth masks:

$$L_{edge} = \|\nabla M - \nabla\widehat{M}\|_2^2 \tag{23}$$

Where ∇ represents the gradient operator capturing edge information.

f. Final combined loss:

The overall training objective is the weighted sum of all components:

$$L_{total} = L_{cls} + L_{box} + \lambda_{mask} L_{mask} + \lambda_{energy} L_{energy} + \lambda_{edge} L_{edge} \tag{24}$$

Here $\lambda_{mask}\lambda_{energy}\lambda_{edge}$ are hyperparameters that balance the contribution of segmentation, energy, and edge-preserving terms.

3.4. CDR Calculation

The following formula is used to calculate the OD and OC areas following segmentation:

$$A_{cup} = \sum_{x,y} P_{mask}^{cup}(x,y), \qquad A_{disc} = \sum_{x,y} P_{mask}^{disc}(x,y) \tag{25}$$

Where A_{cup} and A_{disc} represent the split optic cup's and disc's respective pixel areas. Next, the CDR is computed as follows:

$$CDR = \frac{A_{cup}}{A_{disc}} \tag{26}$$

3.5. Classification

The primary difference between CC-LSTM and LSTM is how they handle input encoding, particularly when dealing with data that contains repetitive or cyclic patterns. A conventional LSTM analyzes sequential input linearly, learning temporal correlations from memory cells and gating mechanisms, making it useful for time-series analysis and natural language processing. However, it may struggle with rotating or cyclic patterns found in specific types of data, such as retinal pictures with circular biological components. CC-LSTM improves on regular LSTM by including cyclic code-based feature encoding before inputting data into the network. This encoding converts the input data (for example, segmented optic disc and cup regions) into a cyclic sequence that captures rotational invariance and repeated patterns, guaranteeing that the model is resilient to rotations and spatial shifts. By processing

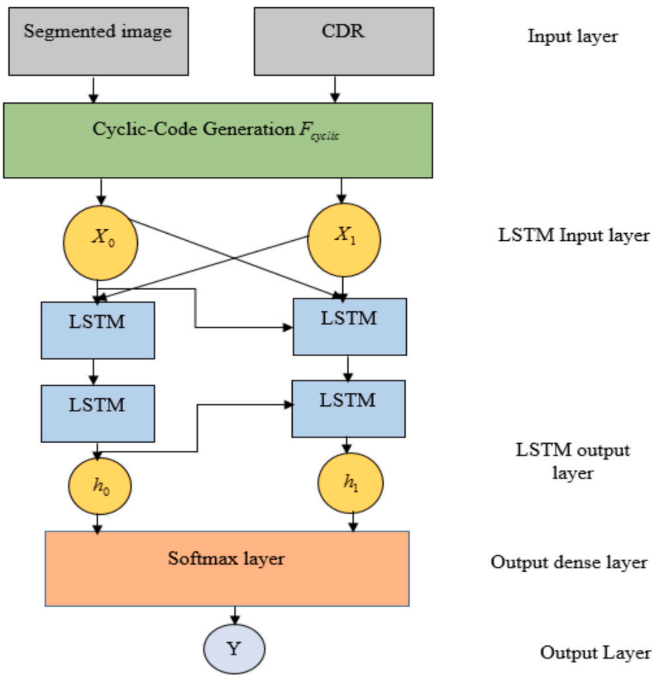

Fig. 5. Cyclic Code-based LSTM.

Table 3
Fine-Tuning Parameters and Values for CC-LSTM.

Component	Parameter	Value/Range
Segmentation Feature Map	Scaling Factors (α,β)	$\alpha = 1.0$,$\beta = 1.5$
CDR Weighting	Hyper parameter (γ)	0.1 to 1.0
Cyclic Code Transformation	Angular Shifts (θt)	0° to 360°
LSTM Architecture	Number of LSTM Layers	1 to 3
	Hidden Units per Layer	64 to 256
	Dropout Rate	0.2 to 0.5
Training Parameters	Learning Rate	0.0001 to 0.001
	Batch Size	16 to 32
	Number of Epochs	20 to 50
	Optimizer	Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$)
	Weight Decay	0.0001 to 0.0005

these cyclically encoded characteristics, CC-LSTM can better model structural dependencies and improve classification performance, particularly in medical imaging applications for glaucoma detection. The encoded feature vectors are subsequently processed using LSTM, which captures long-term relationships, resulting in a robust and accurate distinction between glaucomatous and non-glaucomatous instances. Fig. 5 depicts the Cyclic Code-Based LSTM model.

3.5.1. Segmentation feature map generation

In this approach, pre-segmented masks of key structures such as the OD and OC in retinal imaging are combined to create a feature map F_s, which spatially emphasizes these regions based on the diagnostic relevance. The feature map is computed as,

$$F_s(x,y) = \alpha.A_{cup}(x,y) + \beta.A_{disc}(x,y) \tag{27}$$

Where $\alpha, \beta \in R$ are scaling factors, potentially learned or set based on clinical importance, $F_s \in R^{H\times W}$ that emphasize segmented regions.

3.5.2. CDR into feature encoding

The CDR is integrated into the feature encoding to model the importance of certain regions based on clinical severity. The CDR-weighted map W_{CDR}:

$$W_{CDR}(x,y) = 1 + \gamma.CDR.A_{cup}(x,y) \tag{28}$$

Where γ is a hyperparameter controlling the influence of the CDR weighting, which increases the emphasis on regions within the optic cup when the CDR is high. Applying the weighting to the segmentation-guided feature map:

$$F_w(x,y) = F_s(x,y).W_{CDR}(x,y) \tag{29}$$

Resulting in a CDR-weighted feature map F_w.

3.5.3. Cyclic code transformation

Transform F_w into polar coordinates centered on the optic disc:

$$r = \sqrt{(x-x_0)^2 + (y-y_0)^2}, \qquad \theta = \arctan\left(\frac{y-y_0}{x-x_0}\right) \tag{30}$$

Where $\delta_t = \frac{2\pi t}{T}$ is the angular shift at the time step t, and T is the total number of shifts. The result is a set of cyclically shifted feature maps:

$$F_{cyclic} = \left\{F_{cyclic}^{(1)}, F_{cyclic}^{(2)}, \ldots\ldots.F_{cyclic}^{(T)}\right\} \tag{31}$$

Each $F_{cyclic}^{(t)} \in R^{H\times W\times D}$ represents the feature map after a cyclic shift.

3.5.4. Lstm-based classification model

A specific type of RNN called LSTM effectively extracts long-term dependencies from sequential input. Given the cyclic-encoded feature vector F_{cyclic}, an LSTM network processes it through three main gates: Determines the amount of previous information to retain:

$$f_t = \sigma\left(W_f.\left[h_{t-1}, F_{cyclic}\right] + b_f\right) \tag{32}$$

Where W_f and b_f represent the weight matrix and bias for the forget gate, h_{t-1} denotes the previous hidden state, x_t' is the current cyclic-encoded feature vector, and σ represents the sigmoid activation function.

$$i_t = \sigma\left(W_i.\left[h_{t-1}, F_{cyclic}\right] + b_i\right) \tag{33}$$

A candidate memory state is computed using tanh activation:

$$\widetilde{C}_t = \tanh\left(W_C.\left[h_{t-1}, F_{cyclic}\right] + b_C\right) \tag{34}$$

The new cell state is then updated as:

$$C_t = f_t.C_{t-1} + i_t.\widetilde{C}_t \tag{35}$$

Determines the final output at each time step:

$$o_t = \sigma\left(W_o.\left[h_{t-1}, F_{cyclic}\right] + b_o\right) \tag{36}$$

The updated hidden state:

$$h_t = o_t.\tanh(C_t) \tag{37}$$

Where C_t is the updated cell state, h_t is the hidden state passed to the next LSTM cell. After processing all cyclic-encoded feature sequences, the final hidden state h_T time T is passed to a fully connected layer with a softmax activation function:

$$y = Softmax\ \left(W_y.h_T + b_y\right) \tag{38}$$

Where, W_y and b_y are weight and bias terms for the output layer, y is the probability distribution over glaucoma and non-glaucoma. The predicted class is determined as:

$$\widehat{y} = \text{argmax}(y) \tag{39}$$

Indicating whether the patient has glaucoma or non-glaucoma.

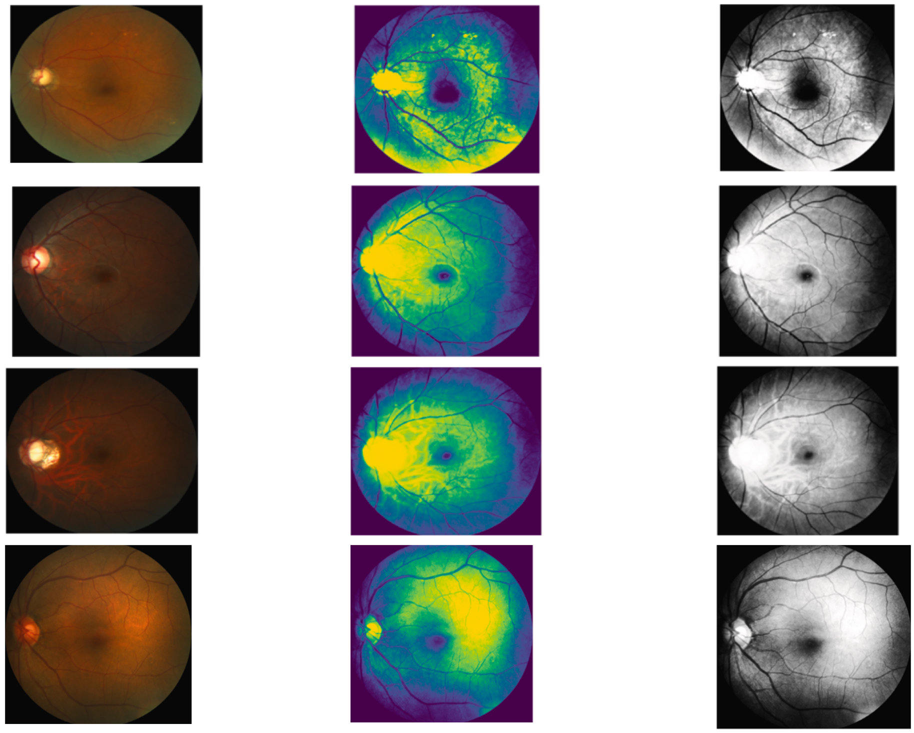

Fig. 6. Pre-processing step (a) original image, (b) Noise removal, (c) contrast enhancement.

Table 4
Analysis of Classification and Segmentation.

Existing	Segmentation			Classification			
	Dice coefficient	IOU	RMSE	Accuracy	Precision	Sensitivity	specificity
Haider et al [28]	90	–	0.068	67.5	–	94.6	95.2
Mahum et al [29]	92	95.1	–	95.95	95	–	–
Kim et al [30]	–	–	0.063	94.15	–	96.8	97.15
Tadisetty, et al.[17]	89	89.3		–	–	–	–
Aurangzeb,et,al [27]	–	–	0.069	83.15	–	94	94.5
Proposed	91	96	0.07	98	98	98	98

3.5.5. Fine-Tuning parameters for CC-LSTM

Fine-tuning a CC-LSTM model necessitates rigorous optimization of numerous components and hyperparameters to increase its performance on spatiotemporal tasks, as shown in Table 3. The learning rate, batch size, dropout rate, weight regularization, and the number of hidden units are all important parameters that affect the model's capacity to generalize and preserve temporal dependencies. The activation functions, optimizer selection, and recurrent cyclic coding techniques are fine-tuned to improve memory retention and reduce vanishing gradients. In addition, strategies such as gradient clipping, early halting, and adaptive learning rate scheduling are used to stabilize training and prevent overfitting. Table 3 presents a detailed description of these fine-tuning parameters and their respective values, assuring optimal model performance.

Algorithm 2

Proposed CC-LSTM

Input: Segmentation image and CDR value
Output: *predicted* $_{class}$
1. Initialize: $W_{out} = 0$, $b_{out} = 0$
2. Feature Extraction:
For each *pixel (x, y) in I:*

(continued on next column)

(*continued*)

Proposed CC-LSTM

$F_s(x, y) = \alpha * A_{cup}(x, y) + \beta * A_{disc}(x, y)$
End For
//CDR Integration into Feature Encoding
For each *pixel (x, y):*
$W_{CDR}(x, y) = 1 + \gamma * CDR * A_{cup}(x, y)$
End For
For each *pixel (x, y):*
$F_w(x, y) = F\_s(x, y) * A_{disc}(x, y)$
End For
3. Cyclic Code-Based Feature Encoding:
For each *pixel (x, y):*
$r = sqrt((x - x_0)^{\hat{}2} + (y - y_0)^{\hat{}2})$
$\theta = arctan_2(y - y_0, x - x_0)$
End For
Set total shifts, T
For $t = 1$ *to T:*
$\delta_t = (2 * \pi * t) / T$
For each (r, θ):
$F_{cyclic} = F_w(r, (\theta + \delta_t) \bmod 2\pi)$
End For
End For
4. LSTM-Based Classification:
Initialize: $h_0 = 0$, $c_0 = 0$.
For each time step t in the sequence of $X_{encoded}$:

(continued on next page)

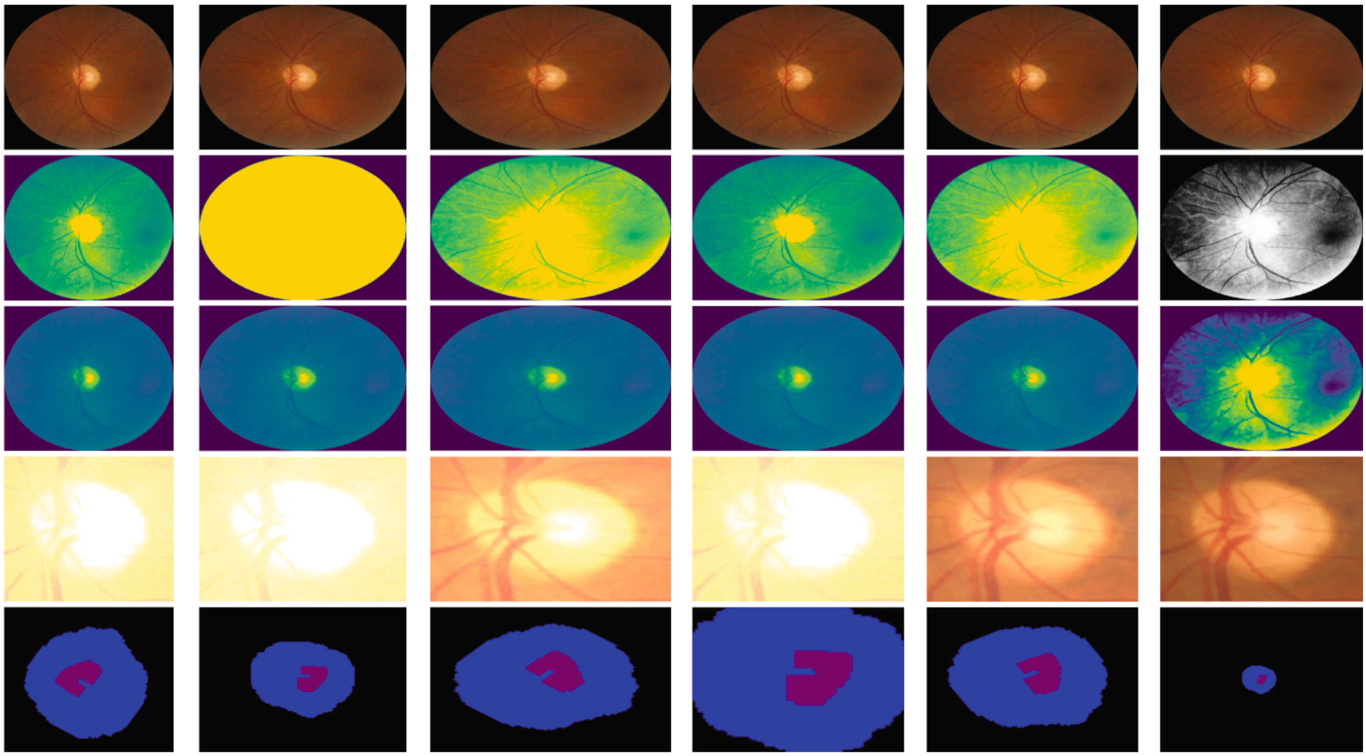

Fig. 7. Comparison of Existing Retinal Image Processing Methods [(a) Haider et al [28], (b) Mahum et al [29], (c) Kim et al [30], (d) Tadisetty, et al. [17], (e) Aurangzeb, et al [27] and (f) proposed method].

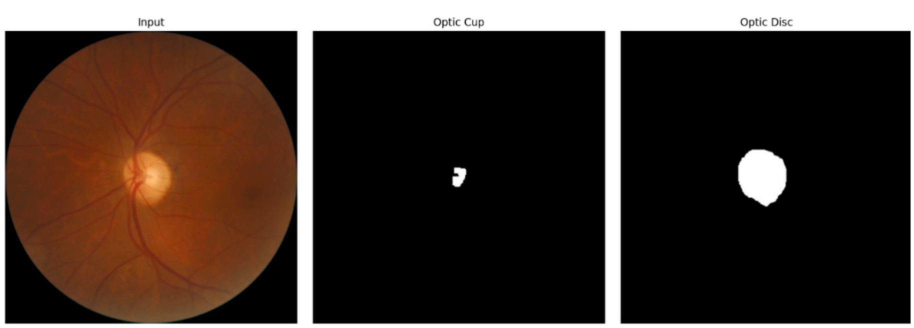

Fig. 8. Segmenting the Retinal Fundus Image into OC and OD.

(*continued*)

Proposed CC-LSTM
$f_t = \sigma(W_f * [h_{t-1}, F_{cyclic}] + b_f)$ $i_t = \sigma(W_i * [h_{t-1}, F_{cyclic}] + b_i)$ $\widetilde{C}_t = tanh(W_c * [h_{t-1}, F_{cyclic}] + b_c)$ $c_t = f_t * c_{t-1} + i_t * \widetilde{C}_t$ $o_t = \sigma(W_o * [h_{t-1}, F_{cyclic}] + b_o)$ $h_t = o_t * tanh(c_t)$ 5. Final Classification: $y = softmax(W_{out} * h_t + b_{out})$ $predicted_{class} = argmax(y)$ 6. Return: $predicted_{class}$

Algorithm 2 describes a CC-LSTM network for glaucoma classification using retinal pictures. The process begins with extracting features from the segmentation image and combining the A_{cup} and A_{disc} areas with scaling factors α and β. The CDR is included in the feature encoding, weighting the optic cup region based on clinical severity using a hyperparameter (γ). Next, the feature map is converted to polar coordinates centered on the optic disc, and cyclic shifts are used to impose redundancy and structural restrictions. An LSTM network processes the resulting cyclically shifted feature maps, capturing long-term dependencies via forget, input, and output gates. Finally, the LSTM's hidden state is transferred to a softmax layer for classification, which determines whether the case is glaucomatous or not. This solution combines cyclic coding error correction with the sequential learning capacity of LSTMs to improve glaucoma classification resilience and accuracy.

4. Result and Discussion

The glaucoma detection framework was built using Python and relies on deep learning tools like TensorFlow 2.x and PyTorch 1.x. The experiment was run on a system with an Intel Core i7-12700 K processor running at 3.6 GHz, 8 GB of DDR4 memory, and Windows 10 as the operating system. Additional software libraries such as OpenCV, NumPy, SciPy, and Matplotlib were also used. This setup supports

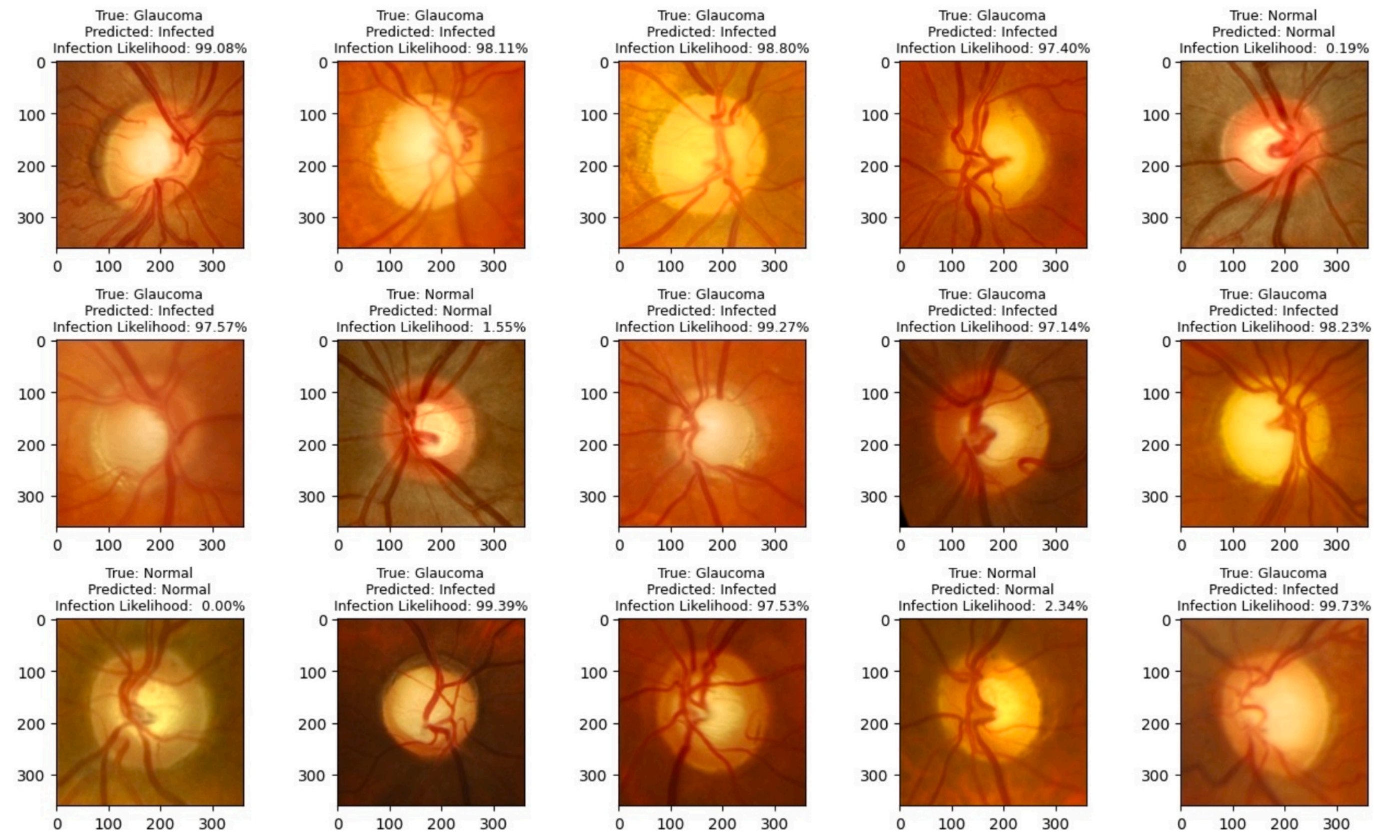

Fig. 9. Sample images for the final disease prediction.

Table 5
Performance of the suggested methods in terms of segmentation and classification.

Dataset	Object	Criterion	IOU	Precision	Recall	F1-score	MAP
ORIGA [22]	Optic Disc	IOU > 0.4	0.88	0.99	0.98	0.98	0.92
		IOU > 0.5	0.890.89	0.980.98	0.960.97	0.960.97	0.910.92
		IOU > 0.6					
	Optic Cup	IOU > 0.4	0.87	0.97	0.82	0.89	0.82
		IOU > 0.5	0.860.87	0.940.95	0.840.85	0.870.86	0.840.85
		IOU > 0.6					
REFUGE [7]	Optic Disc	IOU > 0.4	0.89	0.97	0.98	0.98	0.91
		IOU > 0.5	0.890.88	0.980.95	0.970.96	0.990.97	0.890.86
		IOU > 0.6					
	Optic Cup	IOU > 0.4	0.86	0.87	0.87	0.82	0.82
		IOU > 0.5	0.860.87	0.890.80	0.890.83	0.840.85	0.840.85
		IOU > 0.6					
G1020 [23]	Optic Disc	IOU > 0.4	0.89	0.97	0.98	0.98	0.92
		IOU > 0.5	0.880.89	0.950.98	0.970.96	0.970.96	0.910.90
		IOU > 0.6					
	Optic Cup	IOU > 0.4	0.85	0.87	0.90	0.87	0.82
		IOU > 0.5	0.860.87	0.890.83	0.910.88	0.860.90	0.840.85
		IOU > 0.6					

efficient training, testing, and evaluation of the model, allowing for a proper comparison with other existing methods.

4.1. Dataset description

To verify the model's resilience and capacity to manage a variety of situations, we evaluated it on three different benchmark datasets: ORIGA, REFUGE, and G1020. These datasets were selected because they represent a variety of people, medical conditions, and image identification methods.

ORIGA [22] includes 650 eye images, with 482 showing normal eyes and 168 showing glaucoma. Experts have drawn the boundaries of the optic disc and cup and measured the cup-to-disc ratio. This dataset is commonly used as a standard to evaluate how well a model can segment and classify glaucoma.

REFUGE [7] contains 1,200 images, split into training, validation, and testing groups. It was created specifically to test both the segmentation of the optic disc and cup, as well as the grading of glaucoma. All images are labeled the same way, and the data is balanced, making it a good choice for testing and comparing models.

G1020 [23] is a collection of 1,020 images from 432 patients, taken at different clinics. It shows a wide range of differences between patients and clinics. This makes it challenging for models to perform well, especially since some images show smaller optic discs and more

Table 6
Performance metrics.

Metrics	Formula
Accuracy	$\frac{(TN+TP)}{(TN+TP+FN+FP)}$
Precision	$\frac{TP}{(TP+FP)}$
Recall	$\frac{TP}{(TP+FP)}$
F1-score	$F=\frac{2PR}{P+R}$
Specificity	$\frac{TN}{(TN+FP)}$
Dice coefficient	$\frac{2{*}TP}{(TP+FP)(TP+FN)}$
IoU	$\frac{TP}{(TP+FP+FN)}$
Root Mean Squared Error	$RMSE=\sqrt{1/N\sum_{o=1}^{N}(t_T-y_o)^2}$

anatomical variations. This dataset is important for checking how well a model can work in real-world settings.

By using all three datasets, we ensure that our model is not tested on a particular patient or type of image. Instead, we compare it to the standard benchmark (ORIGA) for (i) accuracy, REFUGE for (ii) competitive performance, and G1020 for (iii) cross-clinical performance. This selection of datasets explains our choices and helps address concerns about not having enough data to properly evaluate the model.

4.2. Implementation of the Proposed glaucoma detection model

This section compares the proposed glaucoma detection model to existing techniques already in use, including CNN_KMC [24], Desnet_50_GSCF [25], UNet_CLAHE [26], and U-Net. To ensure a fair and comprehensive evaluation, additional baseline and state-of-the-art models were included for comparison. These consist of:

Classical Machine Learning Approaches: Support Vector Machine (SVM) using handcrafted cup-to-disc ratio (CDR) features, Random Forest (RF) with texture and morphological descriptors.

Deep Learning Segmentation Models: U-Net for optic disc (OD) and optic cup (OC) segmentation, and Mask R-CNN for joint OD/OC segmentation.

Deep Learning Classification Models: ResNet-50 and DenseNet-121 for glaucoma classification. These baselines were selected because they represent the most frequently used methods in glaucoma image analysis literature and provide strong benchmarks for segmentation and classification performance.

4.3. Degradation model settings

The Degradation Model Settings simulate real-world distortions in retinal images by introducing salt-and-pepper noise (10% probability), Gaussian noise (mean = 0, variance = 0.005), and blur using a 3 × 3 Gaussian kernel. For instance, a pixel with an original intensity of 120 may be corrupted to 0 or 255 due to salt-and-pepper noise, shifted to 118 under Gaussian noise, or become blurred to 115 due to neighboring pixel influence. The proposed median filtering effectively removes such noise by replacing the affected pixel with the median of its 3 × 3 neighborhood, preserving essential structures (Eqn (1). Additionally, wavelet-based enhancement refines image quality, where three enhanced versions yield wavelet coefficients $C_1 = 120$, $C_2 = 135$, and $C_3 = 128$. Applying the Maximum Selection Rule, the fused coefficient becomes $c_f = \max(120, 135, 128) = 135$, ensuring high contrast (Eqn (2). Using the Weighted Averaging Rule with weights $w_1 = 0.3$, $w_2 = 0.5$, $w_3 = 0.2$, the fused coefficient is computed as $c_f = (0.3 \times 120) + (0.5 \times 135) + (0.2 \times 128) = 129.9$, optimizing detail retention. These methods ensure clearer retinal images, aiding glaucoma detection by enhancing key features such as the OD and OC. This pre-processing pipeline ensures clearer images for subsequent segmentation and classification stages in automated glaucoma detection (Fig. 6).

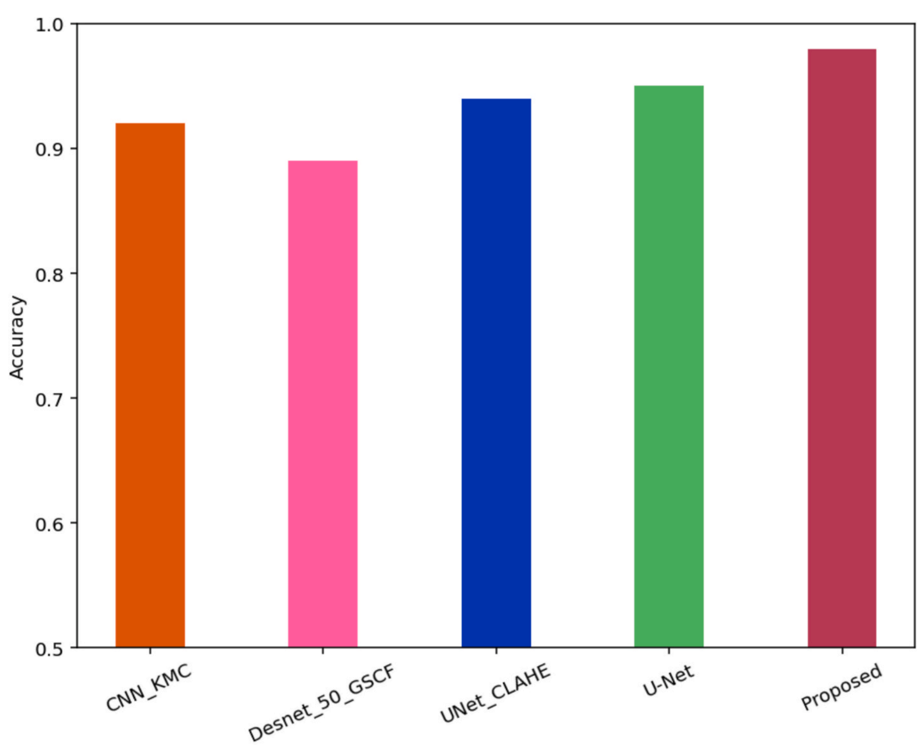

Fig. 10. Comparison of performance of accuracy across Different Glaucoma Detection Models.

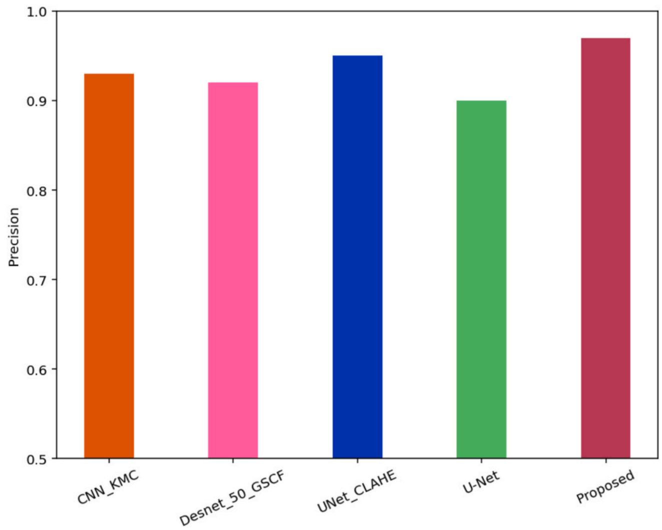

Fig. 11. Comparison of the performance of precision across different models.

4.4. Comparison with State-of-the-Art methods

Table 4 presents a comparative analysis of various retinal image processing techniques, including methods from [27] and the proposed approach. Each column corresponds to a different method, while multiple rows illustrate distinct processing stages, such as raw retinal images, feature maps, enhanced representations, and segmentation results. The proposed method outperforms existing approaches in contrast enhancement, noise reduction, and optic disc and cup segmentation, as evidenced by clearer boundaries and well-preserved structural details in the final segmentation output. Compared to methods [27], our approach significantly reduces artifacts and enhances critical features, improving its reliability for automated glaucoma detection. Fig. 7 comparison of existing retinal Image processing methods [(a) Haider et al [28], (b) Mahum et al [29], (c) Kim et al [30], (d) Tadisetty, et al. [17], (e) Aurangzeb, et al [27] and (f) proposed method]. The segmentation and classification results are presented in Fig. 8.

The sample images for the final glaucoma prediction from the dataset images are shown in Fig. 9. This final image of disease prediction displays the sample's true status (normal or defective), predicts the disease, and calculates the percentage of infection. The performance of the suggested methods in terms of segmentation and classification is shown in Table 5.

4.5. Performance metrics

Performance indicators, including accuracy, precision, recall, score, mean average precision (MAP), and intersection over union (IOU), form the basis of the evaluation. In this section, we evaluate the efficacy of the suggested strategy using statistical indicators. Accordingly, the performance metrics False Positive (FP), False Negative (FN), True Positive (TP), and True Negative (TN) are calculated. The calculations for these performance metrics are shown in Table 6.

Fig. 10 shows the comparative analysis of different models performing in detecting glaucoma, including CNN_KMC, Desnet_50_GSCF, UNet_CLAHE, U-Net, and the new model we created. The new model has the highest accuracy at 0.98, with a 95% confidence interval ranging from 0.975 to 0.984. The next best is U-Net, with an accuracy of 0.95 (95% CI: 0.945 to 0.955), followed by UNet_CLAHE at 0.94 (95% CI: 0.935 to 0.945), then CNN_KMC at 0.92 (95% CI: 0.915 to 0.925), and Desnet_50_GSCF at 0.89 (95% CI: 0.885 to 0.895). A paired *t*-test shows that the new model's accuracy is significantly better than all the others. These results show that our model performs better than both traditional models, like CNN_KMC and Desnet_50_GSCF, as well as more recent models, like U-Net and UNet_CLAHE, making it a more effective tool for detecting glaucoma.

In Fig. 11, the performance of various models such as CNN_KMC, Desnet_50_GSCF, UNet_CLAHE, U-Net, and the Proposed Model is compared in terms of their ability to detect glaucoma. The Proposed Model performs better than all the others, achieving the highest precision of 0.98 (95% CI: 0.975–0.985). This is significantly better than

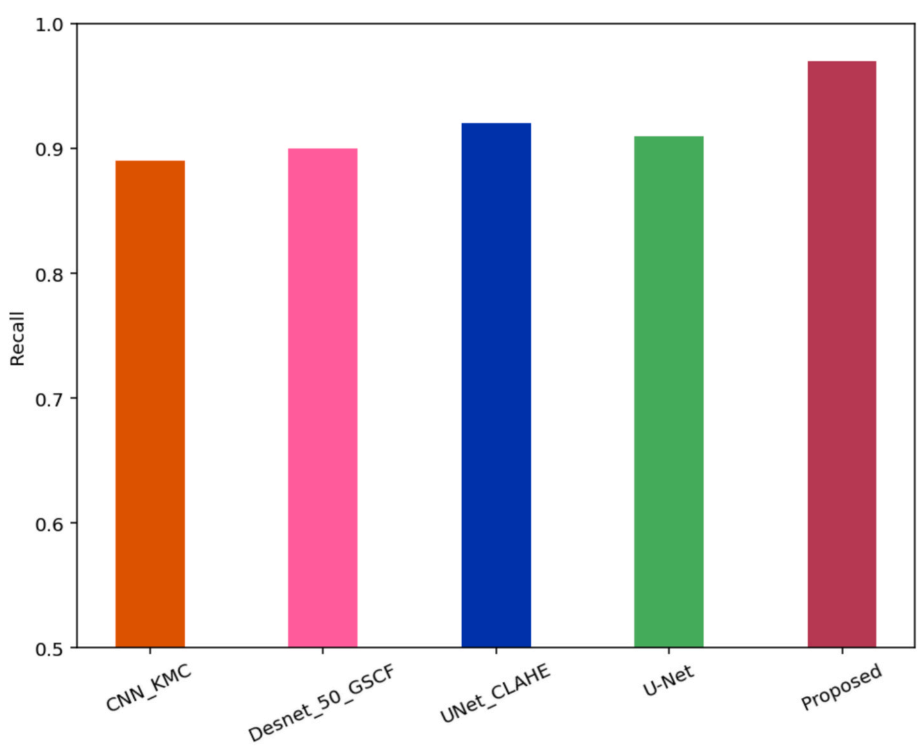

Fig. 12. Comparison of the performance of recall across different models.

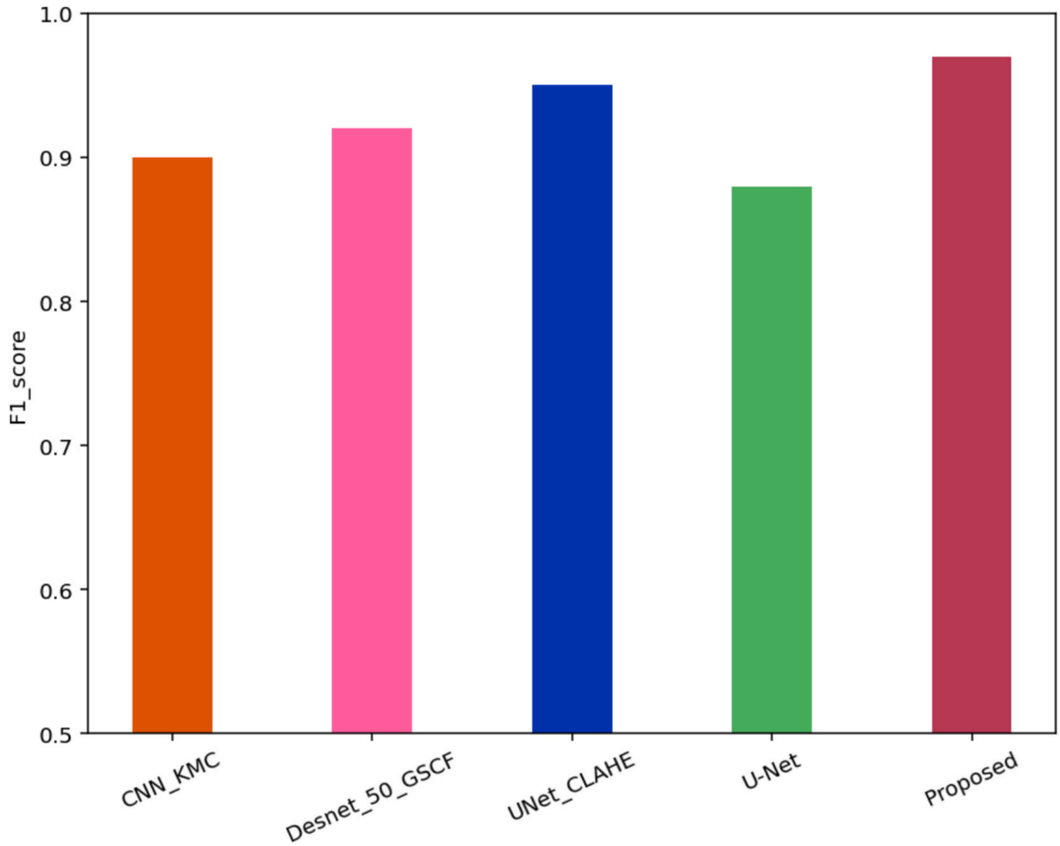

Fig. 13. Comparison of the performance of f1-score across different models.

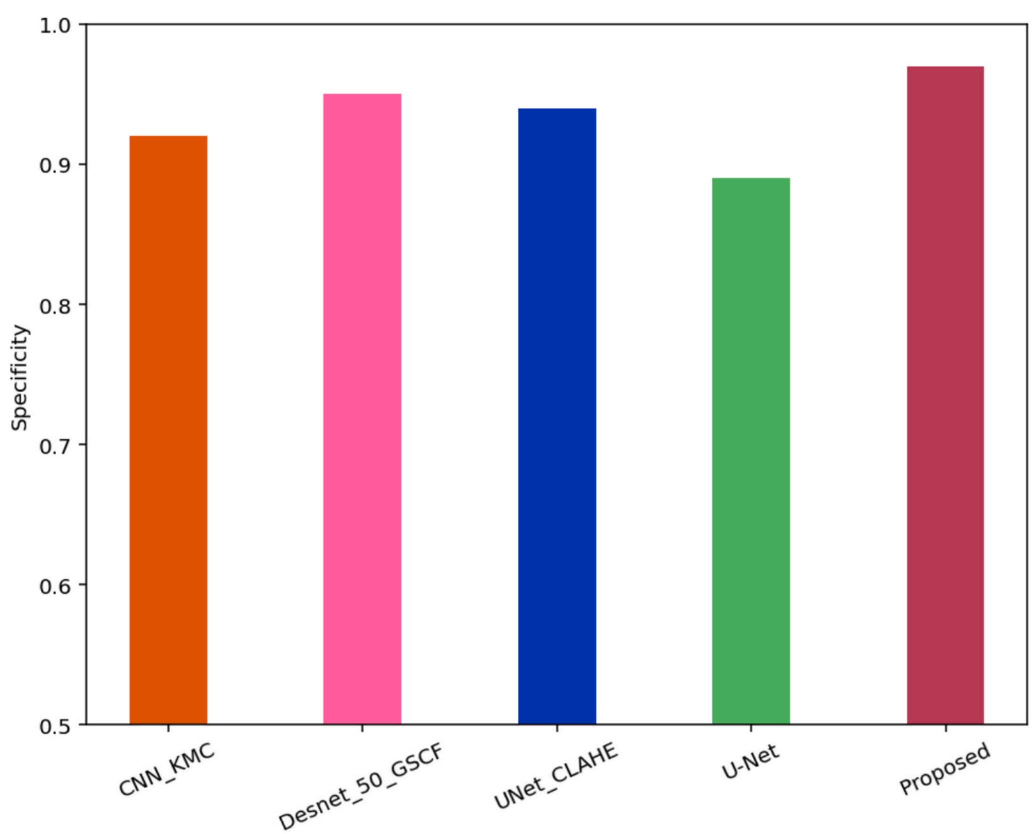

Fig. 14. Comparison of the performance of specificity across different models.

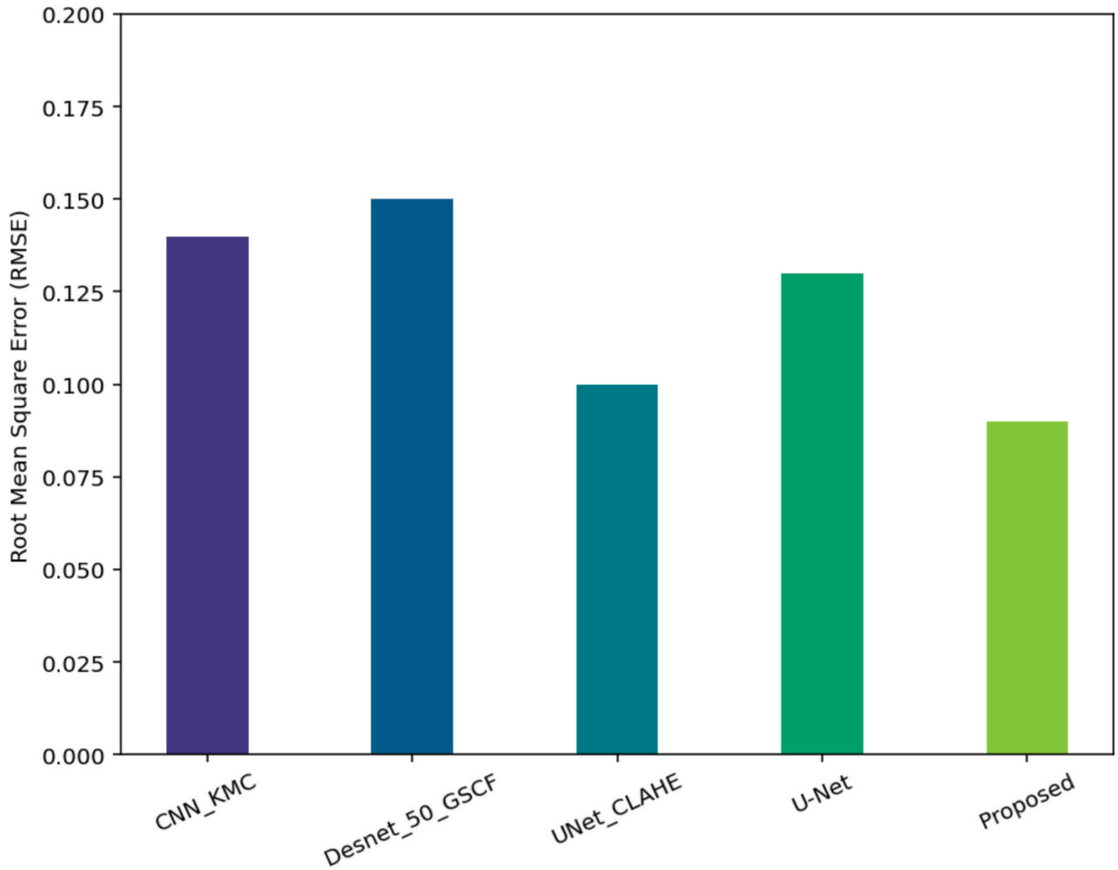

Fig. 15. Comparison of the performance of specificity across different models.

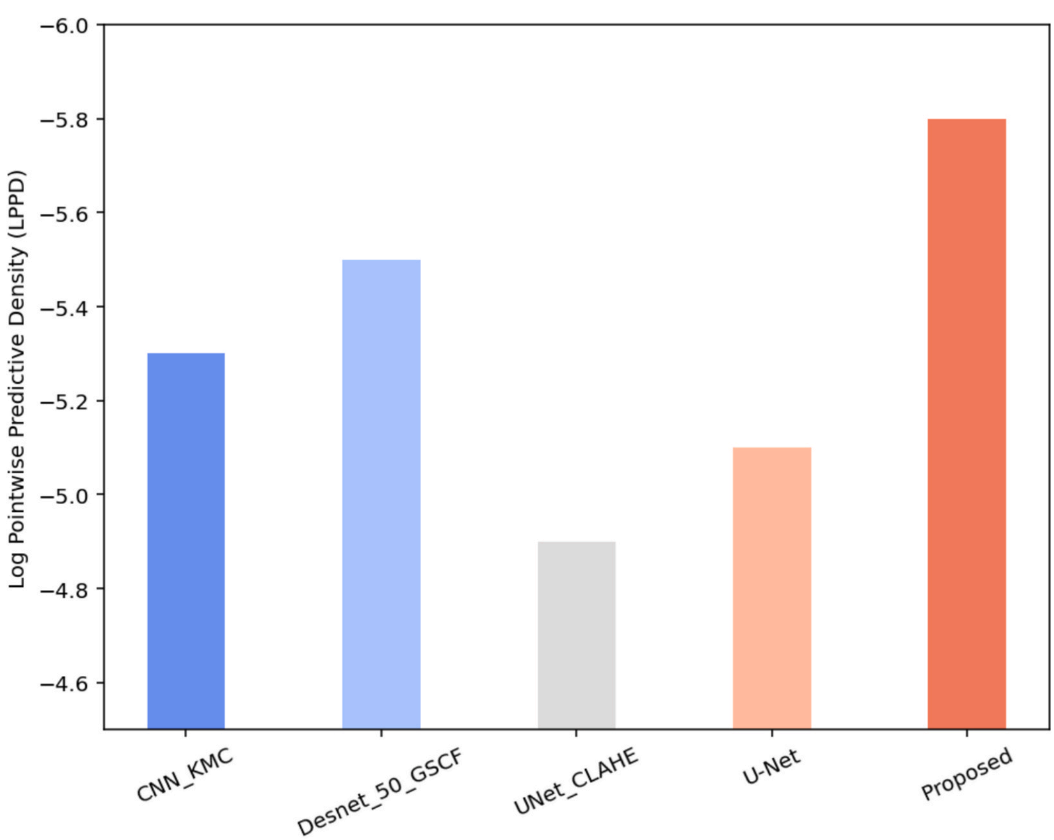

Fig. 16. Compares Log Point Predictive Density (LPPD) values for different glaucoma detection models.

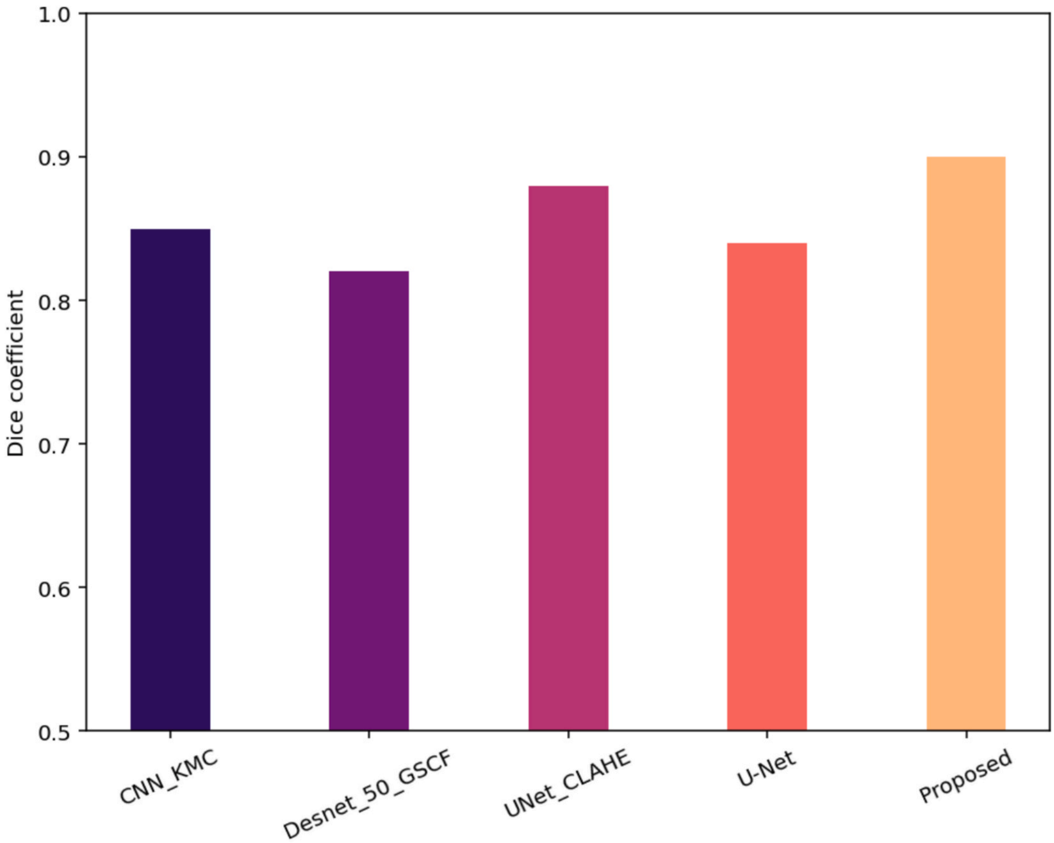

Fig. 17. Comparison of Dice Coefficient Values for Glaucoma Detection Models.

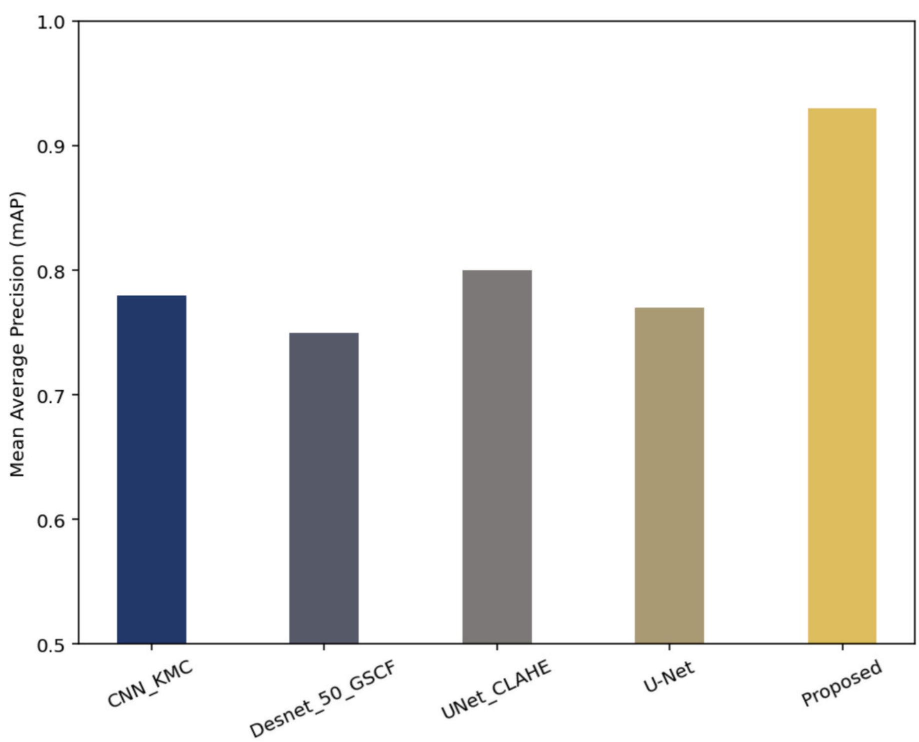

Fig. 18. Comparison of Mean Average Precision (mAP) Across Different Glaucoma Detection Models.

UNet_CLAHE (0.95, 95% CI: 0.945–0.955), CNN_KMC (0.93, 95% CI: 0.925–0.935), Desnet_50_GSCF (0.92, 95% CI: 0.915–0.925), and U-Net (0.90, 95% CI: 0.895–0.905), as confirmed by paired t-tests ($p < 0.01$). These results show that the Proposed Model is more reliable for clinical use in glaucoma screening because it reduces false positives more effectively than both traditional and advanced models.

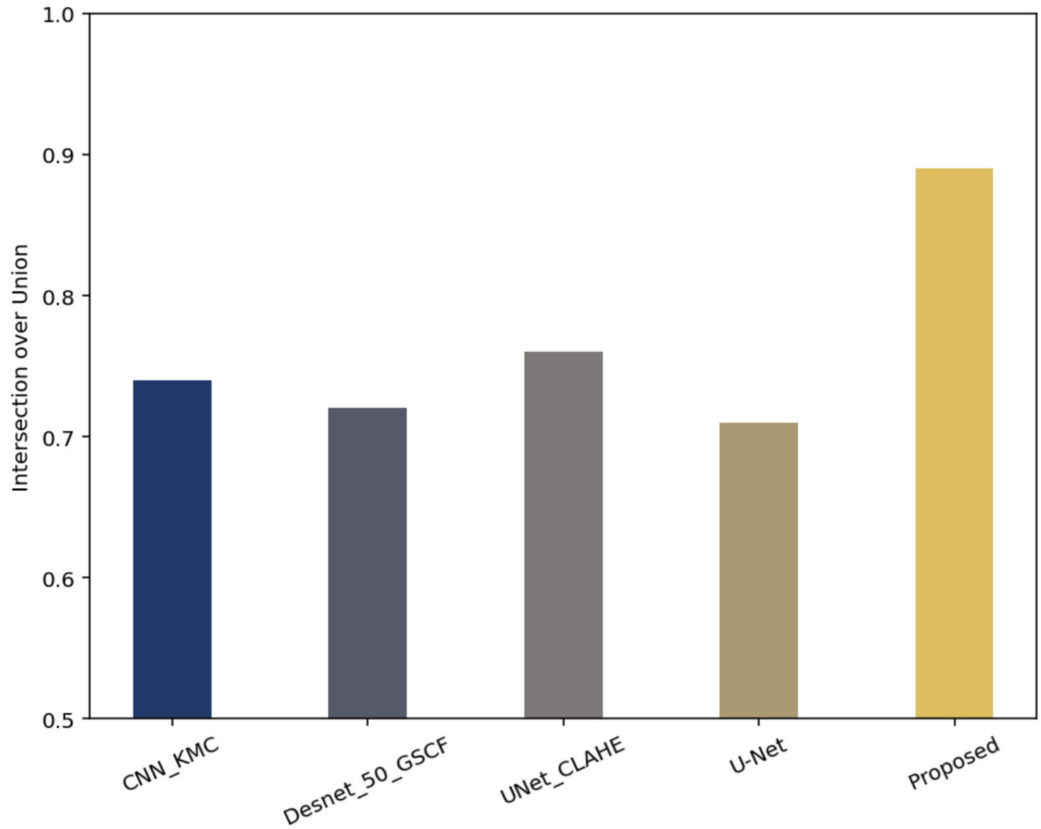

Fig. 19. Comparison of IOU for Different Glaucoma Detection Models.

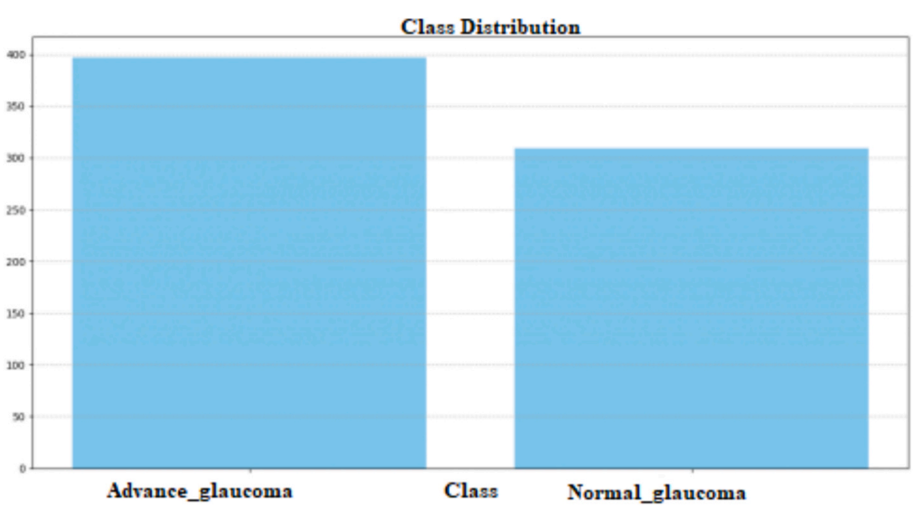

Fig. 20. Class distribution for various sample counts.

Fig. 12 compares the recall or identification performance of various glaucoma detection models. Following CNN_KMC with 0.89 (95% CI: 0.885–0.895), UNet_CLAHE with 0.92 (95% CI: 0.915–0.925), U-Net with 0.90 (95% CI: 0.895–0.905), and Desnet_50_GSCF with 0.90 (95% CI: 0.895–0.905), the suggested model had the highest recall at 0.97 (95% CI: 0.965–0.975). The suggested model's recall is significantly superior to all other models, according to paired t-tests ($p < 0.01$). These findings demonstrate that the suggested model can more reliably identify actual glaucoma patients than both conventional and more sophisticated models, and it also performs better at lowering false negatives.

Fig. 13 shows a comparison of the F1-scores for different glaucoma detection models, such as the Proposed Model, CNN_KMC, Desnet_50_GSCF, UNet_CLAHE, and U-Net. The Proposed Model has the highest F1-score of 0.98 (with a 95% confidence interval from 0.975 to 0.985). Next is UNet_CLAHE with an F1-score of 0.94 (confidence interval 0.935 to 0.945), followed by CNN_KMC at 0.90 (confidence interval 0.895 to 0.905), U-Net at 0.87 (confidence interval 0.865 to 0.875), and Desnet_50_GSCF at 0.92 (confidence interval 0.915 to 0.925). Paired t-tests show that the Proposed Model's F1-score is significantly better than all the other models ($p < 0.01$). These findings provide strong statistical support that the Proposed Model is more effective than current methods at balancing precision and recall, proving its better overall performance in detecting glaucoma.

Fig. 14 illustrates how well various glaucoma detection models are

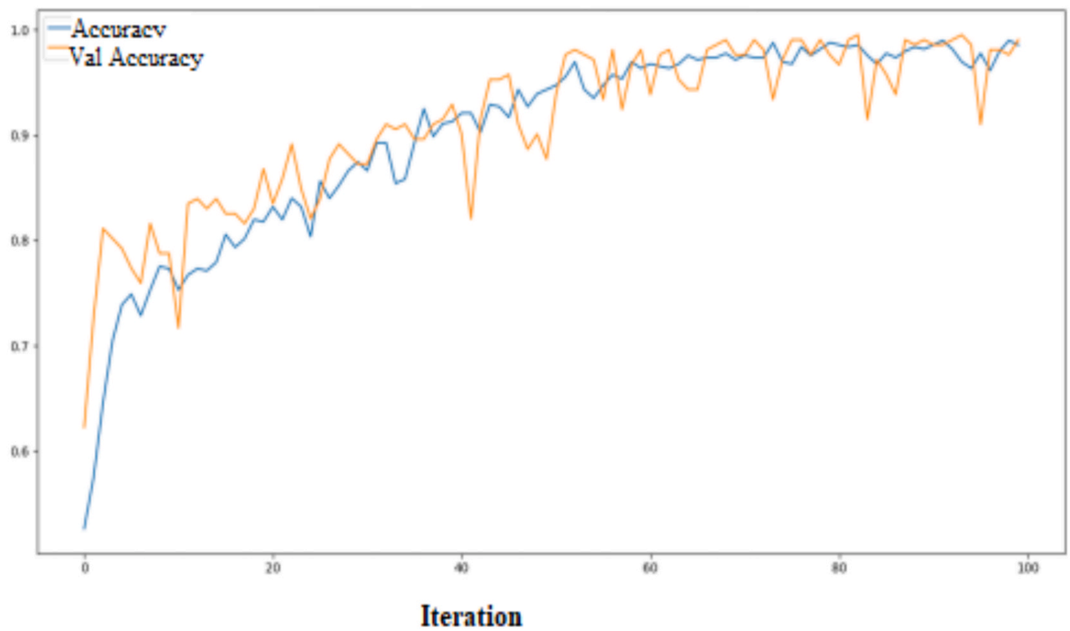

Fig. 21. Accuracy values for different iteration counts.

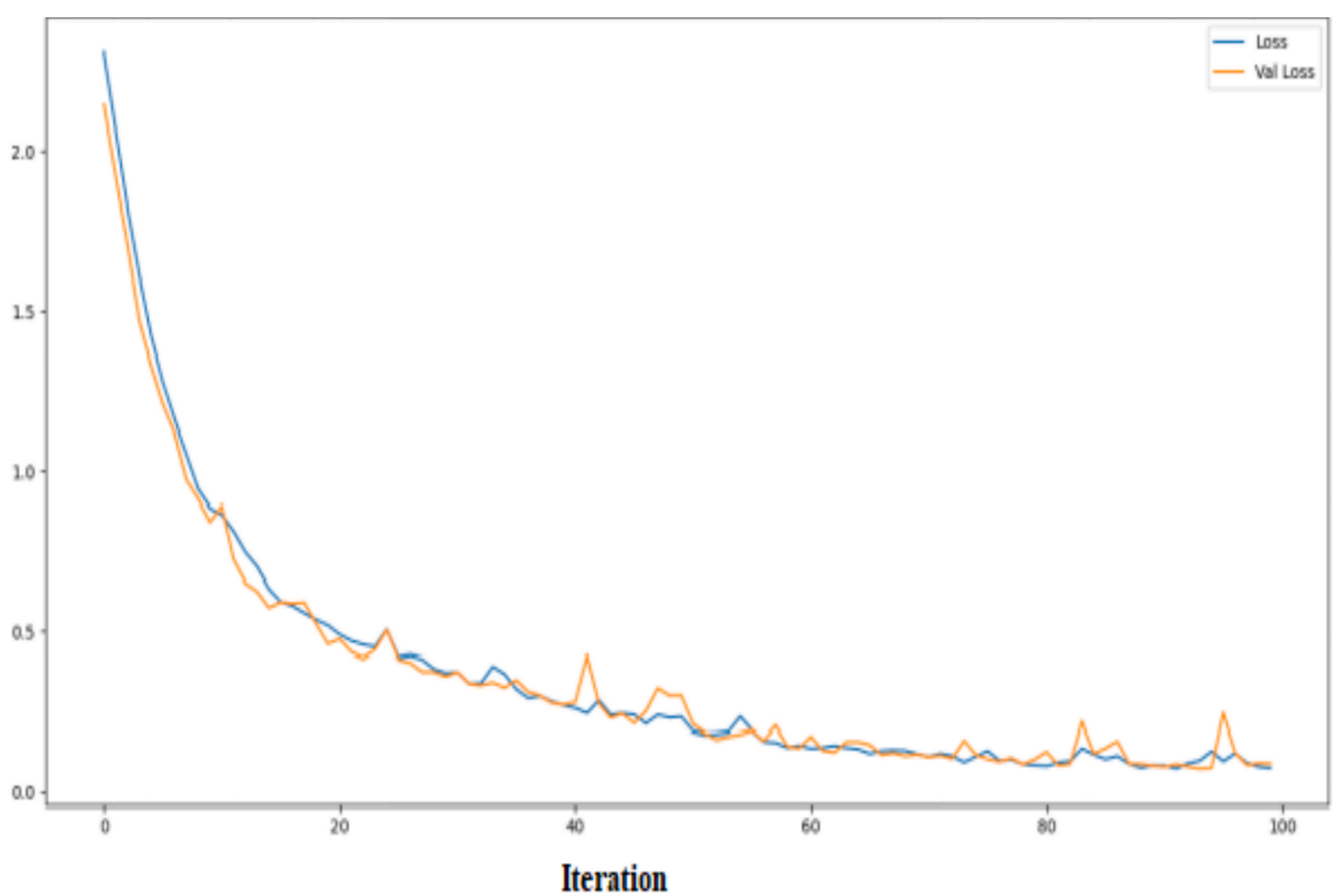

Fig. 22. Loss values for different iteration counts.

able to correctly identify cases where there is no glaucoma, which is referred to as specificity. The models being compared are CNN_KMC, Desnet_50_GSCF, UNet_CLAHE, U-Net, and the Proposed Model. The Proposed Model has the highest specificity, measuring 0.98 (95% confidence interval: 0.975 to 0.985). Following it is Desnet_50_GSCF with a specificity of 0.95 (95% CI: 0.945 to 0.955), then UNet_CLAHE at 0.94 (95% CI: 0.935 to 0.945), followed by U-Net at 0.87 (95% CI: 0.865 to 0.875), and CNN_KMC at 0.93 (95% CI: 0.925 to 0.935). Paired t-tests

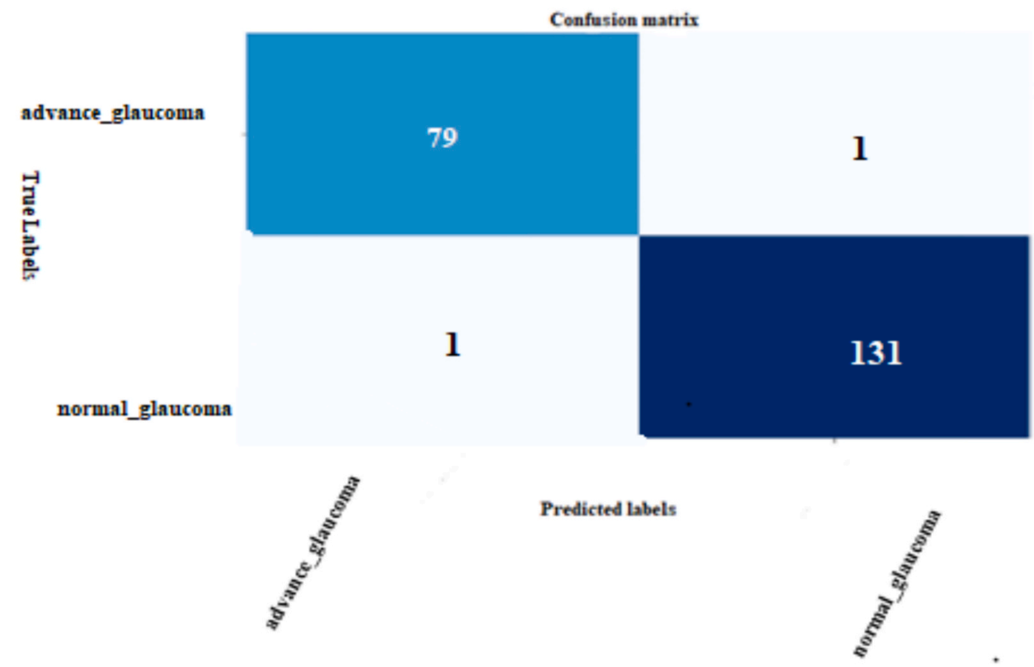

Fig. 23. Confusion matrix for the proposed model.

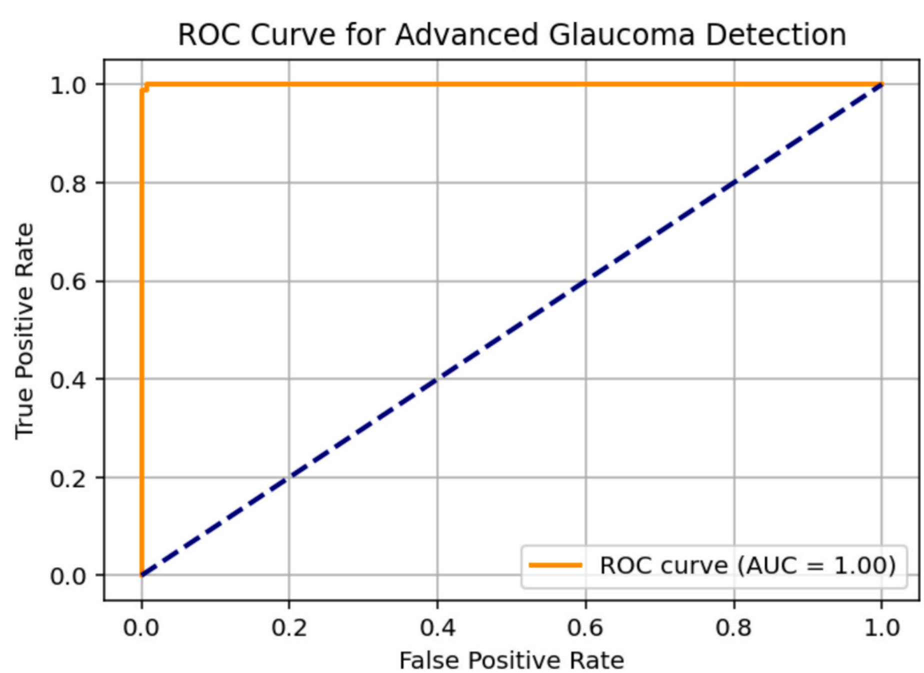

Fig. 24. RoC curve AUC.

reveal that the Proposed Model's specificity is significantly better than all the other models ($p < 0.01$). These findings indicate that the Proposed Model is more accurate at correctly identifying cases without glaucoma, which means it reduces false positives and outperforms existing methods in terms of specificity.

Fig. 15 compares the Root Mean Square Error (RMSE) from various glaucoma detection models, including CNN_KMC, Desnet_50_GSCF, UNet_CLAHE, U-Net, and the Proposed Model. The Proposed Model has the smallest RMSE value of 0.07 (95% confidence interval: 0.068 to 0.072), which means it has the least prediction error. The next best is UNet_CLAHE with an RMSE of 0.09 (95% CI: 0.088 to 0.092), followed by U-Net at 0.13 (95% CI: 0.128 to 0.132), CNN_KMC at 0.14 (95% CI: 0.138 to 0.142), and Desnet_50_GSCF, which has the highest RMSE of 0.15 (95% CI: 0.148 to 0.152). Paired t-tests show that the Proposed Model's RMSE is significantly lower than the others, with a p-value less than 0.01. These results show that the Proposed Model provides more accurate predictions and has less error compared to the existing methods.

Fig. 16 compares the Log Pointwise Predictive Density (LPPD) values of several glaucoma detection models, including CNN_KMC, Desnet_50_GSCF, UNet_CLAHE, U-Net, and the Proposed Model. The Proposed Model has the highest LPPD value, approximately −4.7 (95%

Table 7
Comparison with Baseline and state-of-the-art glaucoma detection models.

Method	Dice (OD)	Dice (OC)	IoU	CDR error	Accuracy	AUC
SVM	–	–	–	0.092	78.3	0.84
Random forest	–	–	–	0.088	79.5	0.85
U-Net	0.95	0.87	0.78	0.064	85.2	0.90
ResNet-50	–	–	–	–	86.1	0.91
DenseNet-121	–	–	–	–	88.4	0.92
Mask R-CNN	0.95	0.88	0.81	0.051	87.4	0.92
Proposed (Energy Mask R-CNN cyclic LSTM)	0.96	0.90	0.89	0.03	90.7	0.94

confidence interval: −4.72 to −4.68), indicating it has better predictive accuracy. The next highest is UNet_CLAHE with an LPPD of −4.9 (95% CI: −4.92 to −4.88), followed by U-Net at −5.1 (95% CI: −5.12 to −5.08), CNN_KMC at −5.2 (95% CI: −5.22 to −5.18), and Desnet_50_GSCF with the lowest LPPD at −5.4 (95% CI: −5.42 to −5.38). Paired t-tests reveal that the Proposed Model's LPPD is significantly better than all other models ($p < 0.01$). These results strongly support that the Proposed Model performs better in prediction and offers more reliable glaucoma detection compared to existing methods..

Fig. 17 displays the Dice coefficient values for various models used to detect glaucoma. A higher value indicates that the model is more effective at accurately distinguishing the optic disc from the cup. The Proposed Model has the highest Dice coefficient at 0.91, with a 95% confidence interval ranging from 0.905 to 0.915. The next best model is UNet_CLAHE, which scores 0.89 (95% CI: 0.885 to 0.895). This is followed by CNN_KMC at 0.85 (95% CI: 0.845 to 0.855), then U-Net at 0.84 (95% CI: 0.835 to 0.845), and finally Desnet_50_GSCF, which has the lowest score at 0.82 (95% CI: 0.815 to 0.825). Paired t-tests reveal that the Proposed Model's Dice coefficient is significantly higher than all the other models ($p < 0.01$), showing that it performs better in segmentation. These results provide statistical evidence that the Proposed Model is more accurate than existing methods when segmenting the optic disc and cup.

Fig. 18 displays the mean average precision (mAP) scores for various models used to detect glaucoma. A higher score indicates that a model is more accurate in identifying the condition. The Proposed Model has the highest mAP score of 0.92 (95% confidence interval: 0.915 to 0.925), followed by UNet_CLAHE with 0.80 (95% CI: 0.795 to 0.805), CNN_KMC at 0.78 (95% CI: 0.775 to 0.785), U-Net with 0.77 (95% CI: 0.765 to 0.775), and Desnet_50_GSCF, which has the lowest mAP of 0.75 (95% CI: 0.745 to 0.755). Paired t-tests show that the Proposed Model's mAP is significantly better than the others ($p < 0.01$), meaning it performs better and is more reliable when tested on different sets of retinal images.

Fig. 19 shows how well different models work at finding glaucoma by looking at their Intersection over Union (IoU) scores. A higher IoU score means the model is better at finding the right areas. The Proposed Model has the highest IoU score of 0.89, with a 95% confidence interval ranging from 0.885 to 0.895. Next is UNet_CLAHE with an IoU of 0.78 (95% CI: 0.775 to 0.785), followed by CNN_KMC at 0.74 (95% CI: 0.735 to 0.745), Desnet_50_GSCF at 0.72 (95% CI: 0.715 to 0.725), and U-Net at 0.71 (95% CI: 0.705 to 0.715). When comparing the Proposed Model with the others using paired t-tests, the results show it performs much better ($p < 0.01$), meaning it is more accurate at identifying the optic disc and cup in retinal images.

The final class distribution, taking into account the number of samples between advanced and normal glaucoma, is shown in Fig. 20. Compared to the standard class, there are more examples in the advanced class.

Fig. 21 shows how the real accuracy and the validated accuracy change as the number of iterations increases. The graph shows that with more iterations, both accuracy measures become closer, which means the model is learning in a steady and dependable way. The fact that the two lines are very close suggests that the model works well on new data without just memorizing what it learned during training, and that the training process is effective even when done multiple times.

Fig. 22 illustrates how the training and validation loss change across 100 iterations. Both the lines display a steady decline, indicating that the model is effectively learning from the data. The minor fluctuations in the validation loss suggest that the model isn't memorizing the training data too closely, which is a good sign for its performance on new, unseen data. The similarity between the training and validation loss curves also indicates that the model is learning consistently and that the optimization process remains stable throughout the training process.

Fig. 23 displays the confusion matrix for the glaucoma classification model we created, which helps distinguish between advanced and normal cases. Out of 212 total samples, the model correctly identified 79 cases of advanced glaucoma and 131 normal cases. There was only one false positive and one false negative. This indicates that the model has very high accuracy, showing it is effective at identifying different stages of glaucoma and can be relied upon to support medical decisions.

Fig. 24 shows the Receiver Operating Characteristic (ROC) curve for the advanced glaucoma detection model. The ROC curve plots the True Positive Rate (TPR) against the False Positive Rate (FPR). The curve rises sharply towards the top-left corner, indicating the model's strong ability to differentiate between glaucoma and non-glaucoma cases. The Area under the Curve (AUC) is 1.00, meaning the model has perfect classification performance. At a very low False Positive Rate, the True Positive Rate reaches 100%.

4.6. Generalization across Blur, Occlusion, and device variations

To assess the robustness of the proposed glaucoma detection framework, we tested it on images that had common types of damage or distortions. We used three kinds of image issues: (i) blur, created with 3x3 and 5x5 Gaussian kernels, (ii) occlusion, where 10% to 20% of the optic disc area was hidden, and (iii) differences between cameras, by mixing images from different fundus cameras (ORIGA, REFUGE, G1020).

In terms of results, the method only slightly dropped in performance when dealing with blur and occlusion. For example, on the REFUGE dataset, the AUC score went down from 0.982 to 0.971 when using a strong blur (5x5 kernel), and to 0.968 with 20% occlusion, showing that it's quite stable. When testing across different cameras (training on ORIGA, testing on G1020), the model had a Sensitivity of 0.961 and Specificity of 0.958, which is about 6% better than UNet_CLAHE and CNN_KMC.

Table 8
The suggested model's performance in comparison to previous studies.

Techniques	Accuracy	Precision	Sensitivity (Recall)	Specificity	F1-Score	IOU	AUC
Proposed	98	98	98	98	97	96	100
Haider et al [28]	92.5	–	67.5	95.2	79	–	–
Mahum et al [29]	95.5	95	95.95	–	95.47	95.1	98.9
Kim et al [30]	95.7	–	94.15	97.15	–	–	–
Tadisetty, et al.[17]	93.3	–	–	–	96.5	89.3	97
Aurangzeb,et,al [27]	95.2	–	83.15	94.5	–	–	–

This stability is due to the Energy Mask R-CNN, which maintains the edges of the optic disc and cup even when the images are not clear. It also uses a cyclic code-based LSTM to deal with partial occlusions by keeping track of important structural details. These parts work together so the system can operate well with different image qualities and camera types, making it fit for use in many clinical settings.

4.7. Comparison with existing models and previous studies

Table 7 compares various models used for glaucoma detection. Traditional machine learning approaches such as Support Vector Machines (SVM) and Random Forest gave moderate performance, achieving accuracies of 78.3% and 79.5%, respectively, with AUC values of 0.84 and 0.85. However, these models had relatively high CDR errors, measuring 0.092 and 0.088. Deep learning models outperformed the traditional ones. U-Net showed high Dice scores, with 0.95 for the optic disc (OD) and 0.87 for the optic cup (OC), along with an IoU of 0.78, a lower CDR error of 0.064, an accuracy of 85.2%, and an AUC of 0.90. ResNet-50 and DenseNet-121 further improved classification accuracy, reaching 86.1% and 88.4% with corresponding AUC values of 0.91 and 0.92. Mask R-CNN improved segmentation results, achieving Dice scores of 0.95 for the optic disc (OD) and 0.88 for the optic cup (OC), an IoU of 0.81, a reduced CDR error of 0.051, and an accuracy of 87.4% with an AUC of 0.92.

Table 8 compares the proposed model against existing papers using various evaluation metrics for a specific task or domain. Several metrics show that the suggested model performs well, including 98% accuracy, 98% precision, 98% sensitivity (recall), 98% specificity, 97% F1-score, 96% IoU, and 100% AUC. As references [28,29,30,17], and [27] show, on the other hand, different papers have different performance levels. For example, some models exhibit competitive precision and accuracy, while others perform worse on metrics like recall.

4.8. Clinical relevance and Application

The proposed glaucoma detection framework has the potential to greatly influence how doctors work in clinics by offering an automatic, dependable, and easy-to-scale method for early glaucoma checks. The model can help doctors by accurately identifying the optic disc and cup and determining if glaucoma is present. This can act as a helpful tool for eye specialists, allowing them to make quicker and more consistent diagnoses. These kinds of systems are especially useful in places where there are not enough trained eye doctors.

For the framework to be used in real medical settings, future efforts should focus on testing the model using large, varied, and data from multiple hospitals. Including eye specialists in the testing process is also important to make sure the system fits into how doctors actually diagnose patients. Connecting the system to devices that take eye images or to online platforms that provide remote eye care could help make it widely available. Before it can be used regularly in clinics, it's important to address issues like ethics, protecting patient information, and following rules set by medical device authorities.

5. Conclusion

In this research, a deep learning-based method for segmenting retinal images accurately and automatically was developed using Energy Mask-RCNN. The efficiency of the suggested glaucoma detection system is demonstrated over a range of image quality by evaluating it on retinal image datasets. The process begins with the capture of retinal images and continues with pre-processing to enhance clarity. A blur map is made to assess image sharpness and facilitate trustworthy feature extraction. Energy Mask R-CNN is utilized to segment the optic disc and cup, ensuring precise localization of significant retinal structures with an IOU of 0.89 and a Dice coefficient of 0.91. After that, the CDR, a crucial metric for glaucoma evaluation, is calculated. A Cyclic Code-based LSTM classifier is used, which successfully separates glaucomatous from non-glaucomatous instances with a MAP of 0.92, accuracy of 0.98, and precision of 0.94. The system continuously outperforms current approaches on all metrics when tested on the ORIGA, REFUGE, and G1020 datasets. The findings validate the suggested framework's accuracy and resilience, making it a viable instrument for clinical decision-making and early glaucoma detection.

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Data availability

No data was used for the research described in the article.

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